RMS Value Of A Periodic Function Of Time:

Let $v(t)$ represent a periodic function of voltage with time period T then we define rms value of voltage function as:

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2\,dt}$$

similarly we can find rms value of current functions or any other physical variable. Mathematically it represents square root of the average of the square of the function.

→ rms values are especially useful for alternating functions with average value being zero, because their average value gives no information about the amplitude of the function. For instance if $v = V_0 \sin\omega t$, $\langle v \rangle = 0$.

Rectified Average Value Of A Periodic Function Of Time:

Suppose $v(t)$ represents a voltage function varying with a time period T then we define its rectified average value as:

$$V_{rav} = \frac{1}{T}\int_0^T |v|\,dt$$

Rectified average value is also sometimes used to characterise physical quantities which have an average value zero.

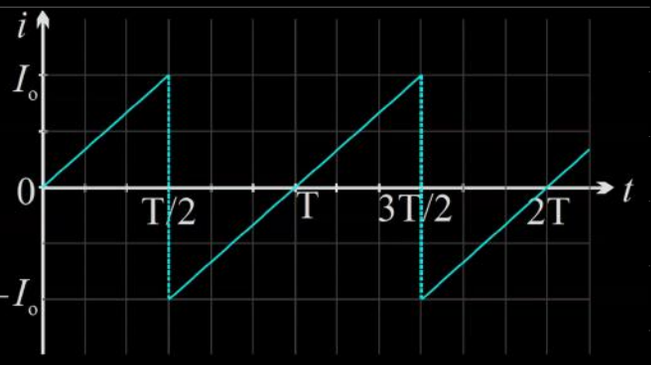
→ Rms and Rav for any function and its mod are same i.e $V_{rms} = |V|_{rms}$ and $V_{rav} = |V|_{rav}$. Because whenever we square or take a mod any negative quantity becomes positive.

→ Rms and Rav of a constant function is that function itself.

→ Ratio of Rms Value to Rav value is called form factor.

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RMS OF SAWTOOTH WAVE:



Consider the sawtooth wave function as shown in the adjacent figure. Even though the period of the function is T, we can see that rms from 0 to T/2 is same as that from T/2 to T due to symmetry. So we can also find the rms considering a time $0 \to T/2$.

Now $i = \frac{2I_0 t}{T}$ (between 0 and T/2). Thus:

$$i_{rms} = \sqrt{\frac{1}{T/2}\int_0^{T/2}\left(\frac{2I_0 t}{T}\right)^2 dt} = I_0/\sqrt{3} \text{ Ans.}$$

→ Average value of t^n from 0 to T is $\left(T^n/_{n+1}\right)$ we could use this result directly:

$i^2 = \frac{4I_0^2}{T^2}t^2 \Rightarrow \langle i^2\rangle$ from 0 to T/2 : $\langle i^2\rangle = \frac{4I_0^2}{T^2}\frac{(T/2)^2}{3}$

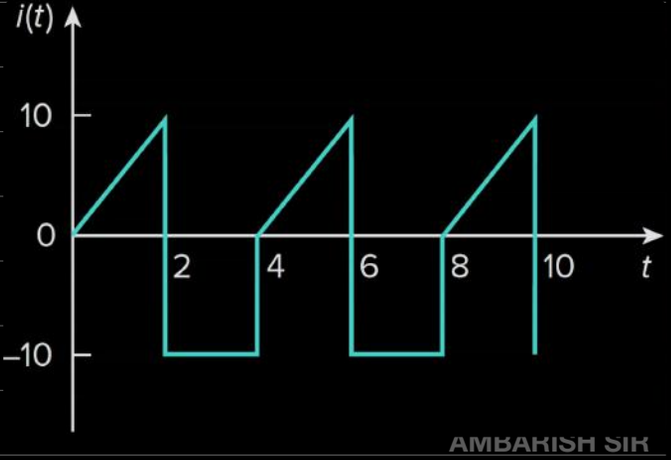
$= I_0^2/3 \Rightarrow i_{rms} = \sqrt{\langle i^2\rangle} = I_0/\sqrt{3}$.

→ If the time period T is further divided in to two piecewise functions of durations T_1 and T_2 such that $T_1 + T_2 = T$. Let i_{1rms} be rms for T_1 and i_{2rms} be rms for T_2 then the net rms is simply obtained as:

$$i_{rms} = \sqrt{\left(T_1 i_{rms_1}^2 + T_2 i_{rms_2}^2\right)/(T_1 + T_2)}$$

EXAMPLE:

Find rms current for the waveform shown in the adjacent figure.



Solution: For rising part $i_{rms} = 10/\sqrt{3}$

For flat parts, $i_{rms} = 10$

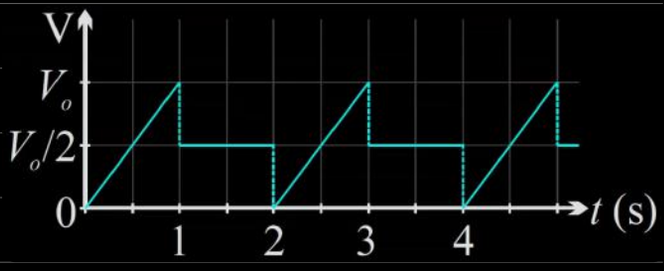
Thus $i_{rms} = \left[\left[\left(\frac{10}{\sqrt{3}}\right)^2 \times 2 + (10^2 \times 2)\right]/4\right]^{1/2}$

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$\Rightarrow \quad i_{rms} = (200/3)^{1/2} = 8.16\ A$ Ans.

EXAMPLE:

Find rms voltage for the shown adjacent graph.

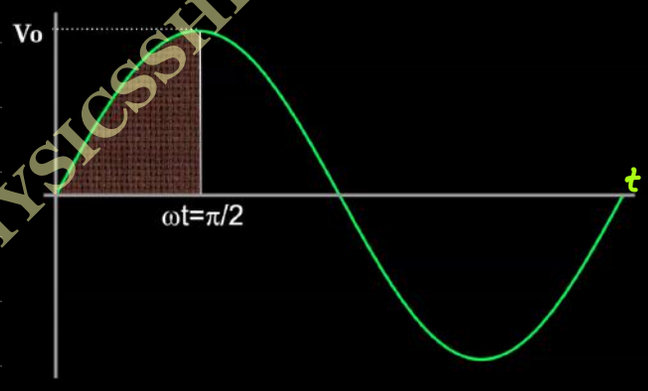


Solution: $V_{rms} = \left[\left(\left(\frac{V_o}{\sqrt{3}}\right)^2 \times 1 + \left(\frac{V_o}{2}\right)^2 \times 1\right)/2\right]^{1/2}$

$\Rightarrow V_{rms} = \sqrt{\frac{7}{24}}\, V_o$

RMS VALUE OF A SINUSOIDAL FUNCTION:

EXAMPLE: Find rms voltage for shown quarter cycle of the voltage function $V = V_0 \sin\omega t$, hence also find rms for the entire cycle.



Solution: Entire cycle can be formed using four pieces of the shaded area, thus the rms of the shaded area is same as rms of the cycle.

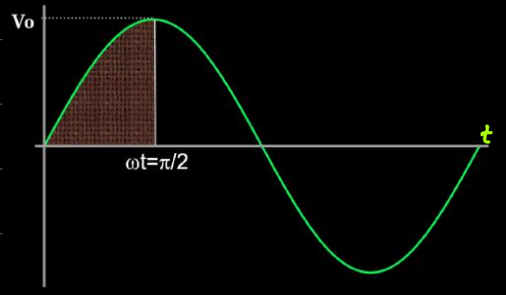
Now for shaded area, $V_{rms} = \left[\frac{2\omega}{\pi}\int_0^{\pi/2\omega} (V_0 \sin\omega t)^2\, dt\right]^{1/2} = V_0/\sqrt{2}$

Thus $V_{rms} = V_0/\sqrt{2}$ Ans. (Both for shaded part as well as entire cycle).

→ For any sinusoidal function of the form $V = V_0 \sin(\omega t + \phi)$ or $V = V_0 \cos(\omega t + \phi)$, $V_{rms} = V_0/\sqrt{2}$.

RECTIFIED AVERAGE VALUE OF SINUSOIDAL FUNCTIONS:

EXAMPLE: Find rectified average value of the shaded portion and hence find rectified average value for the sine function.



Solution: $V_{rav} = \frac{2\omega}{\pi}\int_0^{\pi/2\omega} V_0 \sin\omega t\, dt$

$\Rightarrow \quad V_{rav} = 2V_0/\pi$ Ans. [Both for shaded part

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and the entire cycle).

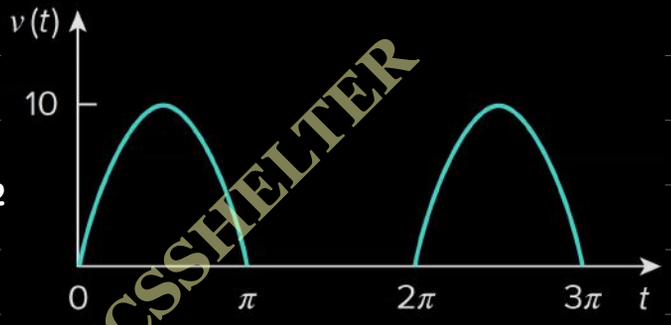
→ Rectified average value of functions $V_0 \sin(\omega t + \phi)$ or $V_0 \cos(\omega t + \phi)$ is $2V_0/\pi \approx 0.64\, V_0$.

→ Rectified average value is smaller than rms for sinusoidal functions.

→ If a time period T can be subdivided in to time durations T_1 and T_2 with rav values v_{rav_1} and v_{rav_2} then net $v_{rav} = \dfrac{(v_{rav_1} \times T_1 + v_{rav_2} \times T_2)}{(T_1 + T_2)}$

EXAMPLE:

Find rms and rav for the adjacent voltage variation.



Solution:

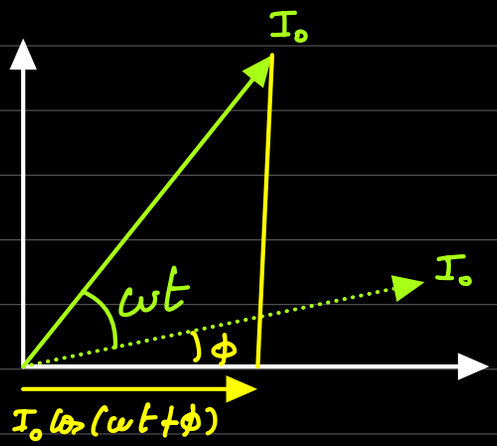
$$v_{rms} = \left[\left(\pi \times (V_0/\sqrt{2})^2 + \pi \times 0 \right) / 2\pi \right]^{1/2}$$

$$\Rightarrow v_{rms} = V_0/2$$

$$v_{rav} = \left[\frac{2V_0}{\pi} \times \pi + 0 \times \pi \right] \frac{1}{2\pi} = \frac{V_0}{\pi} \text{ Ans.}$$

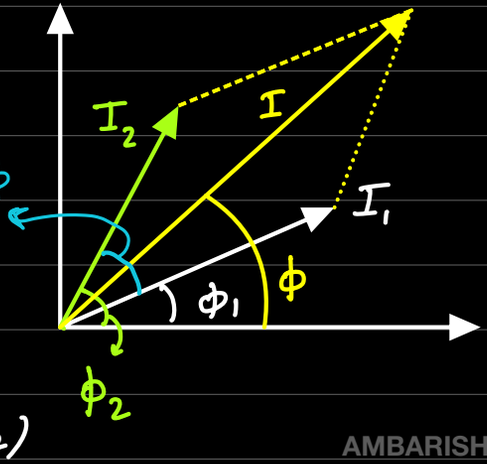
PHASOR ADDITION OF SINUSOIDAL CURRENTS AND VOLTAGES:

Just as during our study of SHM we represented sinusoidal displacements by phasors similarly here we can represent sinusoidal voltages and currents using phasors. For example as current $i = I_0 \cos(\omega t + \phi)$ can be represented as shadow of a rotating vector which started from an angle ϕ with x-axis (see figure). Just like phasor addition in SHM here also we can add the phasors. For instance if $i_1 = I_1 \cos(\omega t + \phi_1)$ and $i_2 = I_2 \cos(\omega t + \phi_2)$ then: $i_1 + i_2 = I \cos(\omega t + \phi)$ where:



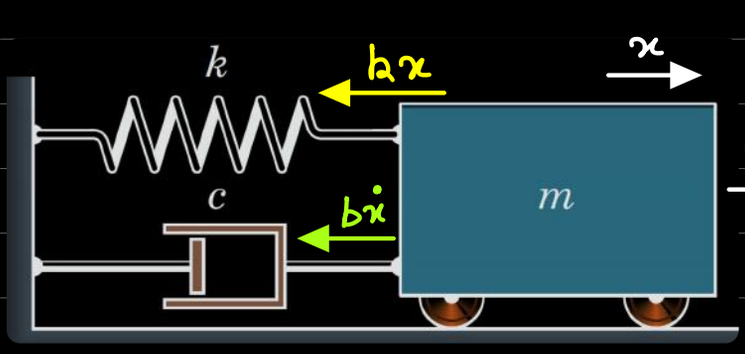
$$I = \left[I_1^2 + I_2^2 + 2 I_1 I_2 \cos(\Delta\phi) \right]^{1/2} \text{ and}$$

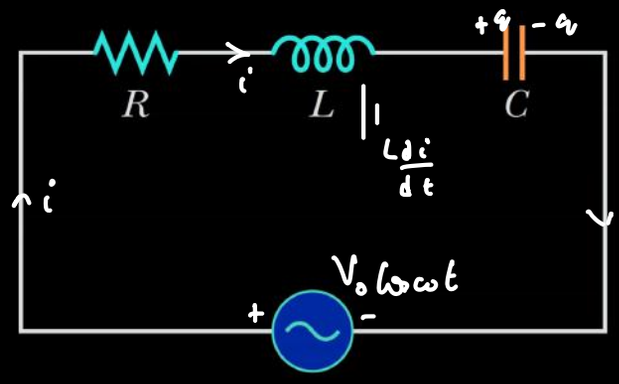
$$\tan\phi = (I_1 \sin\phi_1 + I_2 \sin\phi_2) / (I_1 \cos\phi_1 + I_2 \cos\phi_2)$$



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Electrical Mechanical Analogy For Series LCR Circuits:





Consider the systems shown in the figure above the equation for the left system (as obtained in our study of forced oscillations) is:

$$m\ddot{x} + b\dot{x} + kx = F_0\cos\omega t \quad (1)$$

Now the loop law for the electrical system gives:

$$iR + L\frac{di}{dt} + q/C = \mathcal{E}_0\cos\omega t \quad \text{also } i = dq/dt \text{ thus:}$$

$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = \mathcal{E}_0\cos\omega t \quad (2)$$

We see that the differential equation governing the electrical system has exactly the same form as the mechanical system with following equivalences:

$$L \to m, \quad R \to b, \quad C \to 1/k, \quad \mathcal{E} \to F, \quad i \to v, \quad q \to x$$

Since the equation forms are exactly same therefore we expect the same solution here essentially. (The charge oscillations here follow exactly the same form as the displacement oscillations for the mechanical system). Though here in electrical systems we are generally more interested in the current rather than the charge.

→ Just as in case of forced oscillations initially the system's oscillations are erratic (due to superposition of homogeneous solution over steady state solution) and later on a steady state is achieved, similarly in the electrical system, initially the current is erratic and later follows a steady state oscillatory solution.

→ Both the systems follow a second order linear differential equation.

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Superposition Properties Of Second Order Linear Differential Equations:

1. Let x_H be the solution to the differential equation $a\ddot{x} + b\dot{x} + cx = 0$ and let x_p be the solution to differential equation $a\ddot{x} + b\dot{x} + cx = f(t)$ then $x_p + x_H$ is also a solution to the differential equation $a\ddot{x} + b\dot{x} + cx = f(t)$, x_p is called particular solution and in content of oscillatory electrical/mechanical systems, it represents the steady state solution (recall that x_H has exponentially decaying terms).

2. Let $x_1(t)$ be the solution of the DE $a\ddot{x} + b\dot{x} + cx = f(t)$ and let $x_2(t)$ be the solution of the DE $a\ddot{x} + b\dot{x} + cx = g(t)$ then $x_1(t) + x_2(t)$ is the solution of the DE $a\ddot{x} + b\dot{x} + cx = f(t) + g(t)$
(Here x_1 and x_2 are particular solutions (or steady state solutions).

Euler's Formula:

In this chapter we have seen phasors which are physical quantities of the form $A\cos(\omega t + \phi)$. To represent such quantities we can also use complex numbers about which you will study in great detail in mathematics, however here I will just provide some working knowledge sufficient for our use in solving A.C. circuits. We represent $\sqrt{-1}$ as j, an imaginary number, but still introduction of this number greatly simplifies many real problems as we shall see in this chapter. The first formula that we need here is the euler's formula:

$$e^{j\theta} = \cos\theta + j\sin\theta$$

This formula can be readily verified if you know the series for $\sin\theta$, $\cos\theta$ and e^{θ}. Using the euler's formula, we get an elegant way of expressing sinusoidal quantities, for example suppose $v = V_0\cos(\omega t + \phi)$, we can always write $v = Re(\tilde{v})$ where $\tilde{v}$ is a complex number given as $\tilde{v} = V_0 e^{j(\omega t + \phi)}$ and $Re[\,]$ is an operator which returns the real part of a complex number.

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It can be readily seen from euler's formula that: $\tilde{V} = V_0 e^{j(\omega t+\phi)}$

$\Rightarrow$

$$\tilde{V} = V_0[\cos(\omega t+\phi) + j\sin(\omega t+\phi)], \text{ thus } Re(\tilde{V}) = V_0\cos(\omega t+\phi).$$

How Does Euler Formula Help Solve Second Order D.E.:

Consider the differential equation $m\ddot{x} + c\dot{x} + kx = F(t) = F_0\cos\omega t$ ... (1)
instead of solving this equation we make a similar looking allied equation in another variable $\tilde{x}$ as:

$$m\ddot{\tilde{x}} + c\dot{\tilde{x}} + k\tilde{x} = \tilde{F} = F_0\cos(\omega t) + jF_0\sin\omega t \quad (2)$$

Now RHS of this equation has an extra exciting function which is complex, thus due to superposition, solution of $\tilde{x}$ will have two parts, a real part corresponding to $F_0\cos\omega t$ and a complex part corresponding to $jF_0\sin\omega t$, thus if we can find $\tilde{x}$ we can easily evaluate x as $Re(\tilde{x})$. Now you may still be wondering that all this is fine but still how does it help. So here comes the beauty! We will see that euler's formula reduces differential equations to algebraic equations. Suppose a complex number:

$$\tilde{x} = Xe^{j(\omega t+\delta)} \quad \text{then} \quad \dot{\tilde{x}} = j\omega Xe^{j(\omega t+\delta)} = j\omega\tilde{x}$$
$$\ddot{\tilde{x}} = -\omega^2 Xe^{j(\omega t+\delta)} = -\omega^2\tilde{x} \text{ using these substitutions,}$$

our equation (2) becomes:

$$-m\omega^2\tilde{x} + j\omega c\tilde{x} + k\tilde{x} = \tilde{F} \quad (\text{All derivatives gone!!!})$$

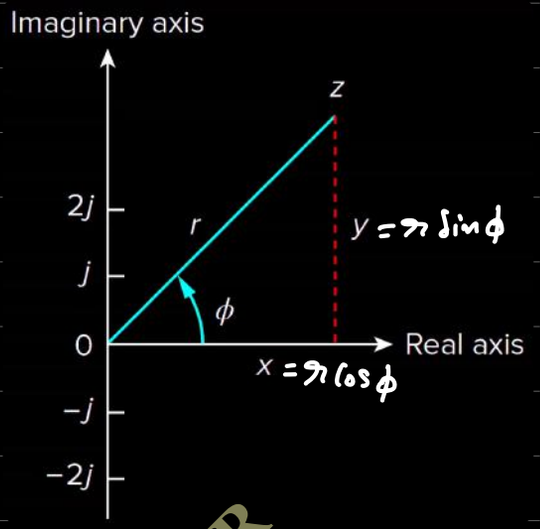
Thus

$$\tilde{x} = \frac{\tilde{F}}{(k-m\omega^2) + j\omega c} \quad \text{and now } x = Re[\tilde{x}].$$

Thus we just needed to solve an algebraic equation instead of a differential equation. So this illustrates the beauty of using complex numbers for solving real problems. To get comfortable in complex number algebra, we will be learning the operations in complex space in the following sections.

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PHASORS AND COMPLEX NUMBERS:



→ A complex number can be represented as $z = x + jy$ where $j = \sqrt{-1}$, x is called the real part and y is called the complex part. Just as we plot vectors, we can plot complex numbers in the "Argand" plane by defining the vertical axis as the complex axis and horizontal axis as the real axis. (see figure).

→ Complex number can also be written in polar or exponential form as $z = r\angle\phi = re^{j\phi}$ (see figure).

→ The conversion from rectangular to polar form is readily inferred from the figure as:

$$r = \sqrt{x^2 + y^2} \quad \text{and} \quad \phi = \tan^{-1}(y/x)$$

→ Addition and subtraction of complex numbers can be done just like vectors. Addition and subtraction are easier in rectangular form while multiplication and division are easier in polar form.

OPERATIONS WITH COMPLEX NUMBERS: ($z_1 = x_1 + jy_1 = r_1\angle\phi_1$ $z_2 = x_2 + jy_2 = r_2\angle\phi_2$)

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle\phi_1 + \phi_2$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\angle\phi_1 - \phi_2$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r}\angle -\phi$$

Square Root:

$$\sqrt{z} = \sqrt{r}\angle\phi/2$$

Complex Conjugate:

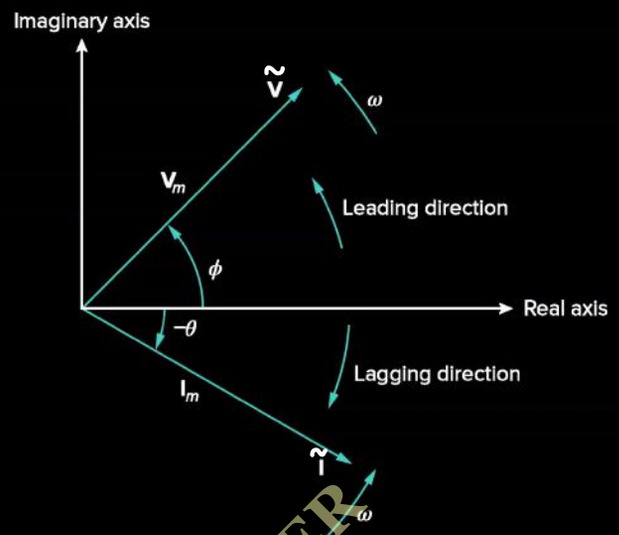
$$z^* = x - jy = r\angle -\phi = re^{-j\phi}$$

Here $z = r\angle\phi$, Also $\frac{1}{j} = -j$

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Leading And Lagging Phasors:

All phasors continuously rotate in the complex plane with an angular velocity ω in the anticlockwise direction. The phasor ahead in the anticlockwise direction is called the leading phasor, for instance in the adjacent figure, $\tilde{V}$ phasor is leading the $\tilde{I}$ phasor.



Representation Equivalents: (Time & Phasor)

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m\cos(\omega t + \phi)$	$V_m\angle\phi$
$V_m\sin(\omega t + \phi)$	$V_m\angle\phi - 90°$
$I_m\cos(\omega t + \theta)$	$I_m\angle\theta$
$I_m\sin(\omega t + \theta)$	$I_m\angle\theta - 90°$

Derivative And Integral Transformations In Phasor Domain:

$$\underbrace{\frac{dv}{dt}}_{\text{(Time domain)}} \Leftrightarrow \underbrace{j\omega\tilde{V}}_{\text{(Phasor domain)}}$$

$$\underbrace{\int v\,dt}_{\text{(Time domain)}} \Leftrightarrow \underbrace{\frac{\tilde{V}}{j\omega}}_{\text{(Phasor domain)}}$$

This can be easily shown, if $\tilde{v} = V_0 e^{j\omega t} \Rightarrow \int \tilde{v}\,dt = \frac{1}{j\omega} V_0 e^{j\omega t}$

$\Rightarrow \int \tilde{v}\,dt = \frac{1}{j\omega}\tilde{v}$ similarly $\frac{d\tilde{v}}{dt} = j\omega V_0 e^{j\omega t} = j\omega\tilde{v}$.

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EXAMPLE: Using the phasor approach determine the current in a circuit governed by integrodiffenential equation:

$$4i + 2\int i\,dt - \frac{di}{dt} = 50\cos(2t + 75^\circ)$$

Solution: $\omega = 2$, let $\tilde{i}$ be the allied phasor then:

$$4\tilde{i} + \frac{2}{j\omega}\tilde{i} - j\omega\tilde{i} = 50\angle 75^\circ$$

$$\Rightarrow \tilde{i}[4 - j - 2j] = 50\angle 75^\circ \Rightarrow \tilde{i} = 50\angle 75^\circ/(4-3j)$$

$$\Rightarrow \tilde{i} = 50\angle 75^\circ / (5\angle -37^\circ) = \frac{50}{5}\angle(75^\circ - (-37^\circ)) = 10\angle 112^\circ$$

Thus $i = 10\cos(2t + 112^\circ)$ Ans.

EXAMPLE: Find the voltage in a circuit described by integrodiffenential equation:

$$2\frac{dv}{dt} + 6v + 10\int v\,dt = 50\cos(5t - 30^\circ)$$

Solution: $\omega = 5$, now let $\tilde{v}$ be the allied phasor then:

$$2j\omega\tilde{v} + 6\tilde{v} + \frac{10}{j\omega}\tilde{v} = 50\angle -30^\circ$$

$$\Rightarrow \tilde{v}[6 + j[10 - 2]] = 50\angle -30^\circ \Rightarrow \tilde{v}[6 + 8j] = 50\angle -30^\circ$$

$$\Rightarrow \tilde{v} = 50\angle -30^\circ/(10\angle 53^\circ) = \frac{50}{10}\angle(-30^\circ - 53^\circ)$$

$$\Rightarrow \tilde{v} = 5\angle -83^\circ \Rightarrow v = 5\cos(5t - 83^\circ)$$ Ans

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Phasor Relationships For Circuit Elements:

Resistor:

Let $i = I_m \cos(\omega t + \phi)$ then,

$$v = iR = I_m R \cos(\omega t + \phi)$$

In phasor form:

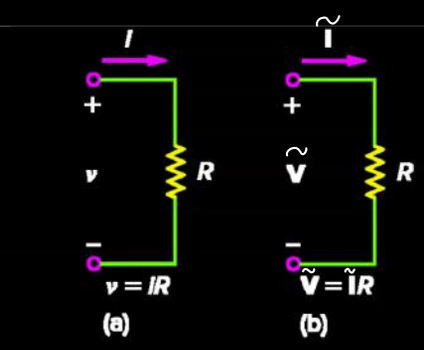
$$\tilde{V} = R I_m \angle \phi$$

and $\tilde{I} = I_m \angle \phi$

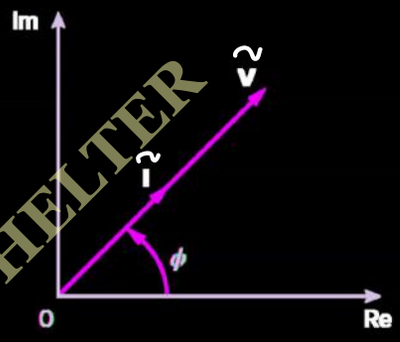
Thus:

$$\boxed{\tilde{V} = \tilde{I} R}$$

Thus in a resistor there is no phase difference between the current phasor and the voltage phasor.



Voltage-current relations for a resistor in the: (a) time domain, (b) frequency domain.



Phasor diagram for the resistor.

Inductor:

Corresponding to a current $i = I_m \cos(\omega t + \phi)$

We have for an inductor,

$$v = L\frac{di}{dt} = -\omega L I_m \sin(\omega t + \phi)$$

$$\Rightarrow v = \omega L I_m \cos(\omega t + \phi + 90^\circ)$$

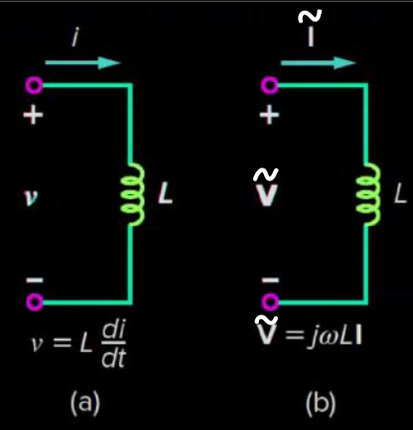
Thus in phasor domain:

$$\tilde{V} = \omega L I_m \angle \phi + 90^\circ$$

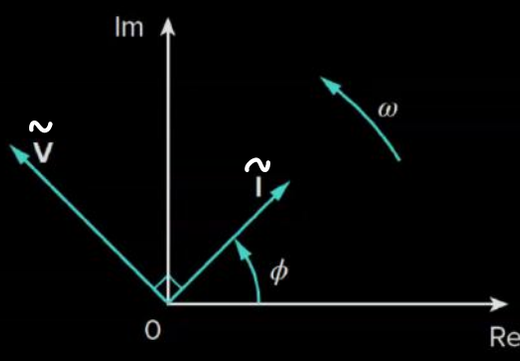
$$\Rightarrow \tilde{V} = \omega L I_m e^{j(\omega t + \phi)} \times e^{j\pi/2}$$

$$\Rightarrow \boxed{\tilde{V} = j\omega L \tilde{I}}$$

Thus for an inductor voltage leads the current by an angle of 90°.



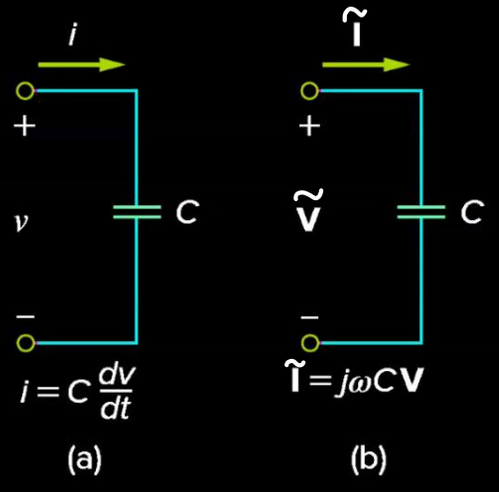
Voltage-current relations for an inductor in the: (a) time domain, (b) frequency domain.



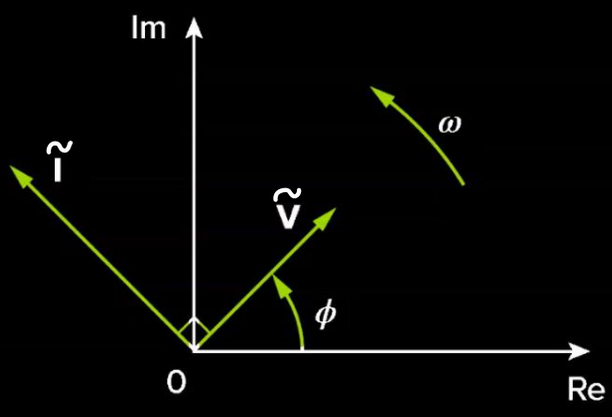
Phasor diagram for the inductor; **I** lags **V**.

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CAPACITOR:



Voltage-current relations for a capacitor in the: (a) time domain, (b) frequency domain.



Phasor diagram for the capacitor; **I** leads **V**.

Let us assume voltage across capacitor to be $v = V_m \cos(\omega t + \phi)$

Now $i = C\,dv/dt = -C\omega V_m \sin(\omega t + \phi) = C\omega V_m \cos(\omega t + \phi + 90^\circ)$

In phasor domain, $\tilde{i} = C\omega V_m e^{j(\omega t + \phi)} \times e^{j\pi/2}$

$\Rightarrow \tilde{i} = j\omega C V_m e^{j(\omega t + \phi)}$, but $\tilde{v} = V_m e^{j(\omega t + \phi)}$ thus,

$\boxed{\tilde{i} = j\omega C \tilde{v}}$, thus in a capacitor current leads the voltage by 90°.

→ The phase relations between current and voltages for an inductor and capacitor can be memonized by the mnemonic "CIVIL"

Capacitor: C: I > V (Phase of current is greater)
Inductor: V > I : L (Phase of voltage is greater).

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Summary of voltage-current relationships.

Element	Time domain	Phasor domain
R	$v = Ri$	$\tilde{\mathbf{V}} = R\tilde{\mathbf{I}}$
L	$v = L\frac{di}{dt}$	$\tilde{\mathbf{V}} = j\omega L\tilde{\mathbf{I}}$
C	$i = C\frac{dv}{dt}$	$\tilde{\mathbf{V}} = \frac{\tilde{\mathbf{I}}}{j\omega C}$

EXAMPLE:

The voltage $v = 12\cos(60t + 45°)$ is applied to a 0.1-H inductor. Find the steady-state current through the inductor.

Solution: $\omega = 60$, $\tilde{V} = 12\angle 45° = j\omega L\tilde{i} \Rightarrow \tilde{i} = \tilde{V}/j\omega L$

$\Rightarrow \tilde{i} = (12\angle 45°)/(6j) = (12\angle 45°)/(6\angle 90°) = 2\angle -45°$

thus $i = 2\cos(60t - 45°)$ Ans.

EXAMPLE: If a voltage $v = 10\cos(100t + 30°)$ V is applied to a 50 μF capacitor, calculate the current through the capacitor.

Solution: $\omega = 100$, $\tilde{V} = 10\angle 30° = \tilde{i} \times \frac{1}{j\omega C} \Rightarrow \tilde{i} = \tilde{V} \times j\omega C$

$= 10\angle 30° \times (100 \times 50 \times 10^{-6})\angle 90° = 50\angle 120°$ mA thus:

$i = 50\cos(100t + 120°)$ mA Ans.

IMPEDANCE AND ADMITTANCE: In a previous section, we obtained the voltage current relations for the three elements as:

$\tilde{V} = \tilde{I}R$, $\tilde{V} = j\omega L\tilde{i}$ and $\tilde{V} = \tilde{i}/j\omega C$

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We can rearrange these equations as:

$\frac{\tilde{v}}{\tilde{i}} = R$, $\frac{\tilde{v}}{\tilde{i}} = j\omega L$ and $\frac{\tilde{v}}{\tilde{i}} = \frac{1}{j\omega C}$, thus we see that a capacitor and inductor behave just like a resistance but a complex one. The ratio of voltage to current across any element is called Impedance ($\tilde{Z}$) and for all purposes, it can be treated just like a resistor. For any circuit branch:

$\tilde{Z} = \tilde{v}/\tilde{i}$ or $\tilde{v} = \tilde{i}\tilde{Z}$, If we know the $\tilde{Z}$ of circuit elements we can apply loop law just as we did in the DC circuits. All DC theorems namely superposition, thevenin, millman, compensation, norton, star-delta, two-port and wheatstone etc are exactly applicable to A.C. circuits also, except that here impedances are complex numbers.

→ For maximum power transfer, external impedance is complex conjugate of internal impedance.

→ Just like D.C. circuits, we can combine impedances according to series and parallel rules and also divide the voltages in ratios of impedances for series connections.

→ Reciprocal of impedance is called Admittance ($\tilde{Y}$). In parallel combinations, currents get divided in ratio of reciprocal impedances (or admittances).

→ Even though $\tilde{v}$ and $\tilde{i}$ are phasors, their ratio $\tilde{v}/\tilde{i}$ is not a phasor, rather it is just a complex constant. This can be easily seen: say $\tilde{v} = v_0 e^{j(\omega t+\phi)}$ and $i = i_0 e^{j(\omega t+\delta)}$ then $\tilde{v}/\tilde{i} = \frac{v_0}{i_0} e^{j(\phi-\delta)} = \tilde{Z}$ thus there is no ωt in $\tilde{Z}$, it is just a complex constant. When we write $\tilde{Z} = Z\angle\theta$ we simply mean $\tilde{Z} = Z e^{j\theta}$ and not $\tilde{Z} = Z e^{j(\omega t+\theta)}$.

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REACTANCE: If a complex impedance is given as $\tilde{Z} = R \pm jx$

then $R = Re(\tilde{Z})$ is known as resistance and $x = Im(\tilde{Z})$ is known as reactance.

→ The impedance $\tilde{Z} = R \pm jx$ can be expressed in polar form as $\tilde{Z} = |Z| \angle \theta$ where $|Z| = \sqrt{R^2 + x^2}$ and $\theta = \tan^{-1}(\pm x/R)$.

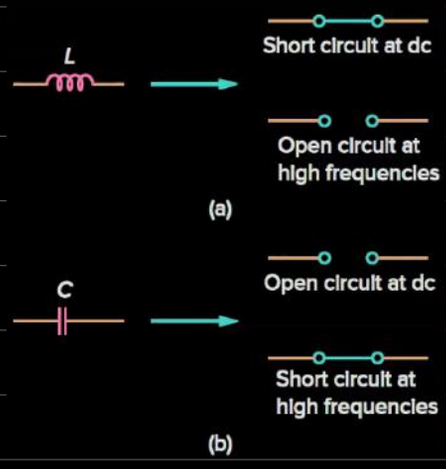
→ Reactance of a capacitor and inductor depend on the driven angular frequency ω thus impedance of any circuit in general will depend on driven ω.

BEHAVIOR OF PASSIVE ELEMENTS (L,R,C) AT VERY HIGH AND VERY LOW FREQUENCIES

The adjacent table lists out impedances and behaviours of L, R, C elements at very low (d.c.) frequencies and very high frequencies. If we let $\omega \to 0$ we can call it a d.c. response. For an inductor, $\tilde{Z} = j\omega L$, so if $\omega = 0$, $\tilde{Z}_L = 0$, thus for very low frequencies, inductor acts almost like a short circuit. At very high frequencies, $|\tilde{Z}_L| \to \infty$, thus at very high frequencies an inductor acts like an open circuit. For a capacitor, $|\tilde{Z}_C| = \frac{1}{\omega C}$ thus at very low frequencies it act like an open circuit and at very high frequencies, it behaves like a short circuit.

Impedances and admittances of passive elements.

Element	Impedance	Admittance
R	$\tilde{Z} = R$	$\tilde{Y} = \frac{1}{R}$
L	$\tilde{Z} = j\omega L$	$\tilde{Y} = \frac{1}{j\omega L}$
C	$\tilde{Z} = \frac{1}{j\omega C}$	$\tilde{Y} = j\omega C$



FINDING ADMITTANCE FROM IMPEDANCE:

Let $\tilde{Z} = R + jx \Rightarrow \tilde{Y} = G + jB = \frac{1}{R + jx} \times \frac{R - jx}{R - jx}$

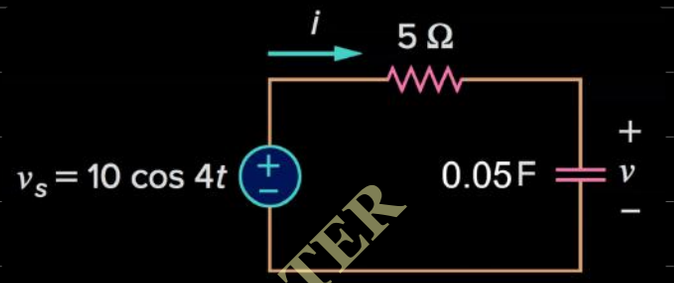
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$\Rightarrow G + jB = \frac{R - jx}{R^2 + x^2}$ thus $G = \frac{R}{R^2+x^2}$ and $B = \frac{-x}{R^2+x^2}$

thus $\tilde{Y} = \tilde{Z}^* / Z^2$ where $\tilde{Z}^*$ represents conjugate of $\tilde{Z}$

EXAMPLE:

Find $i(t)$ and $v(t)$ for the shown circuit.



Solution: Method I: Complex Numbers:

$\omega = 4$, $\tilde{X}_c = 1/j\omega C = -j/\omega C = -j/(4 \times 0.05) = -5j$

$\Rightarrow \tilde{Z} = 5 - 5j$ (∵ series resistor and capacitor). Now:

$\tilde{I} = \tilde{V}_s / \tilde{Z} = 10\angle 0 / (5 - 5j) = 10\angle 0 \times \frac{5+5j}{5^2+5^2} = \frac{10}{50}\angle 0 \times 5\sqrt{2}\angle 45^\circ$

$\Rightarrow \tilde{I} = \sqrt{2}\angle 45^\circ = \sqrt{2}\cos(4t + 45^\circ)$ Amperes. Ans.

Now In series voltage drops are in direct ratio of impedances. Thus,

$$\tilde{V} = \tilde{V}_s \times (\tilde{X}_c / \tilde{Z}) = (10\angle 0) \times [(5\angle -90^\circ)/(5\sqrt{2}\angle -45^\circ)]$$

$\Rightarrow \tilde{V} = 5\sqrt{2}\angle -45^\circ = 5\sqrt{2}\angle -45^\circ$ thus $v = 5\sqrt{2}\cos(4t - 45^\circ)$ Ans.

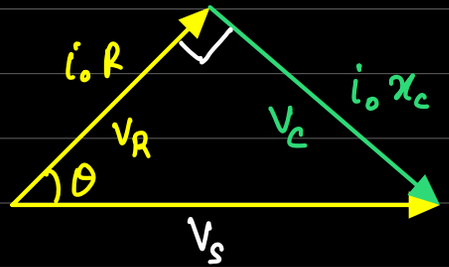
Method 2: Geometrical method using triangle or parallelogram law of addition:

Recall that in a resistor voltage is in same phase as current while in capacitor voltage lags behind the current.

From the figure:

$$i_0\sqrt{X_c^2 + R^2} = V_s$$

$$\Rightarrow i_0 = \frac{V_s}{\sqrt{X_c^2+R^2}} = \frac{10}{\sqrt{5^2+5^2}} = \sqrt{2}\ A$$



Now $\tan\theta = X_c/R = 1 \Rightarrow \theta = 45^\circ$, thus,

$i = \sqrt{2}\cos(4t + 45^\circ)$, Now $V_c = V_s \sin\theta = 10 \sin 45^\circ = 5\sqrt{2}$ Volt.

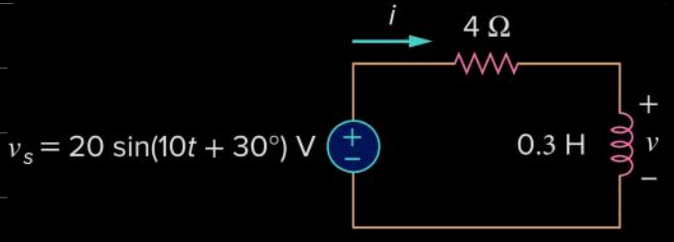
also V_c is 90° behind i thus:

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$V_C : v = 5\sqrt{2}\cos(4t + 45° - 90°) = 5\sqrt{2}\cos(4t - 45°)$ Ans.

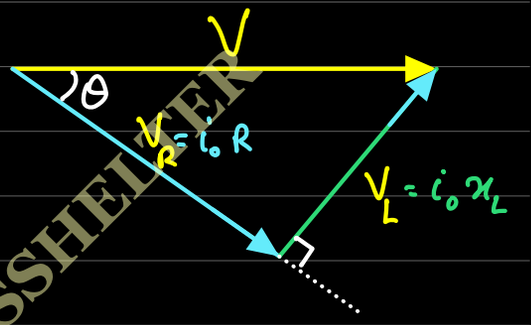
EXAMPLE:

Find $v(t)$ and $i(t)$ for the shown circuit in the adjacent figure.



Solution: Method I: Geometrical Method:

$\omega = 10$, $X_L = \omega L = 3\Omega$. Now in an inductor, voltage is $\pi/2$ ahead of the current. Thus we get the phasor triangle as shown in the figure. Now $V_s = i_0\sqrt{X_L^2 + R^2}$ $\Rightarrow$



$$i = V_S/\sqrt{X_L^2 + R^2} = 20/\sqrt{3^2 + 4^2} = 4\,A$$

Now $\tan\theta = X_L/R = 3/4 \Rightarrow \theta = 37°$, thus current is 37° behind the voltage phasor. Thus, $i = 4\sin(10t + 30° - 37°)$

$\Rightarrow i = 4\sin(10t - 7°)$ Ans.

Now $V_L = V_s\sin\theta = 20 \times \sin 37° = 12$ Volts. Also angle of V_L is 90° ahead of the angle of current, thus,

$$V_L = 12\sin(10t - 7° + 90°) = 12\sin(10t + 83°)$$ Ans.

METHOD 2: USING COMPLEX NUMBERS:

$\omega = 10$, $\tilde{X}_L = j\omega L = 3\hat{j}$ $\Rightarrow$ $\tilde{Z} = 4 + 3\hat{j}$, $\tilde{V} = 20\angle 30° - 90°$

$= 20\angle -60°$

Now $\tilde{i} = \tilde{V}/\tilde{Z} = (20\angle -60°)/(5\angle 37°) = 4\angle -97°$, thus:

$i = 4\cos(10t - 97°) = 4\sin(10t - 7°)$ Ans.

Now:

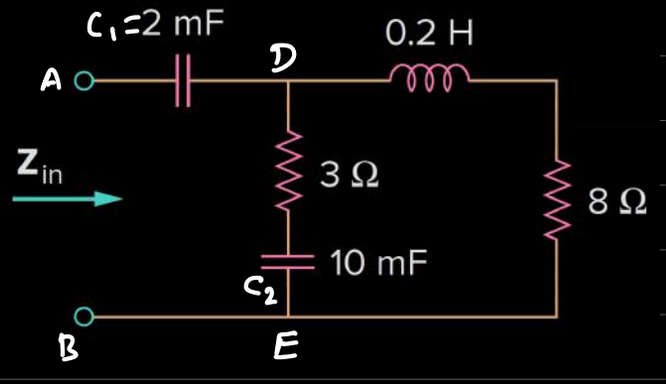
$$\tilde{V}_L = \frac{\tilde{X}_L}{\tilde{Z}} \times \tilde{V}_s = \frac{3\angle 90°}{5\angle 37°} \times 20\angle -60° = 12\angle -7°$$

Thus $V_L = 12\cos(10t - 7°) = 12\sin(10t + 83°)$ Ans.

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EXAMPLE :

Find the effective impedance of the shown circuit if driving frequency is $\omega = 50$ rad s^{-1}!



Solution: $\omega = 50 \Rightarrow \tilde{x}_{C_1} = -j/\omega C = -10j$

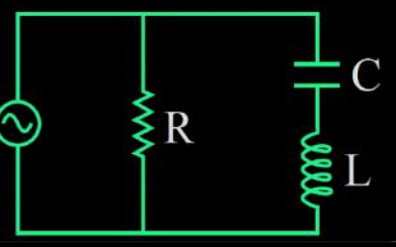
$\tilde{x}_L = j\omega L = 10j$, $\tilde{x}_{C_2} = -2j$

Thus $\tilde{Z}_{eff} = [(8+10j) \,||\, (3-2j)] \text{ Ser } (-10j)$

$$\Rightarrow \tilde{Z}_{eff} = \frac{(8+10j)(3-2j)}{11+8j} - 10j = \frac{(44+14j)(11-8j)}{11^2+8^2} - 10j$$

$$= \frac{596-2048j}{185} = 3.22 - 11.07j \ \Omega \text{ Ans.}$$

EXAMPLE : In the shown circuit, $x_C = 100\,\Omega$, $x_L = 200\,\Omega$ and $R = 100\,\Omega$. Find the current through the source and its rms value.



Solution: $\tilde{x}_L = 200j$ $x_C = -100j \Rightarrow \tilde{Z} = 100 \,||\, [(200-100)j]$

$$\Rightarrow \tilde{Z} = \frac{100 \times 100j}{100+100j} = \frac{100j}{1+j} \frac{(1-j)}{(1-j)} = 50+50j$$

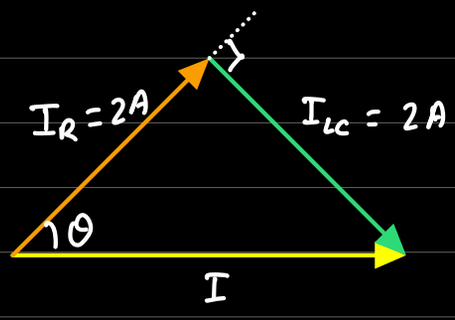
$$\Rightarrow \tilde{Z} = 50\sqrt{2} \angle 45^\circ \Rightarrow \tilde{i} = \tilde{v}/\tilde{Z} = (200 \angle -90^\circ)/(50\sqrt{2} \angle 45^\circ)$$

$$\Rightarrow \tilde{i} = 2\sqrt{2} \angle -135^\circ \Rightarrow i = 2\sqrt{2} \cos(\omega t - 135^\circ) = 2\sqrt{2} \sin(\omega t - 45^\circ)$$

Thus $i_{rms} = i_{max}/\sqrt{2} = 2\,A$ Ans.

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Method II: Geometrical method



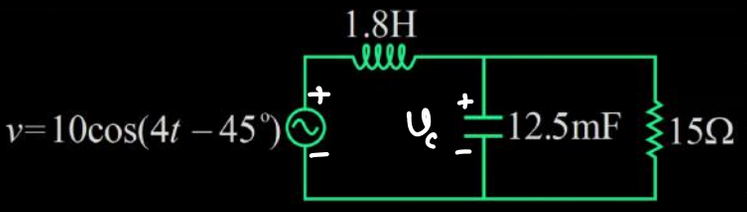
In the resistor, voltage and current are in same phase while in an inductor voltage leads (or current lags behind voltage) by 90°. Let θ be the angle by which total current in the circuit lags behind the voltage. Now $I_R = 200/100 = 2A$, $I_{LC} = 200/(200-100) = 2A$ (overall inductance effect dominates).

Note that voltage is along the orange phasor ((current phasor in resistor) Thus from diagram, $I = \sqrt{2^2+2^2} = 2\sqrt{2}$ Amperes, $\theta = 45°$
Thus net current lags behind the voltage by 45°. Thus:

$$i = 2\sqrt{2}\sin(\omega t - 45°) \quad \text{Ans.}$$

(For parallel circuits we take vector sum of current phasors, for series we take vector sum of voltage phasors).

EXAMPLE: For the shown circuit find v_c as a function of time.



Solution: Using complex numbers:

$x_c = -j/\omega C$, $\omega = 4 \Rightarrow \tilde{x}_c = -20j\ \Omega$, $\tilde{x}_L = j\omega L = 7.2j\ \Omega$.

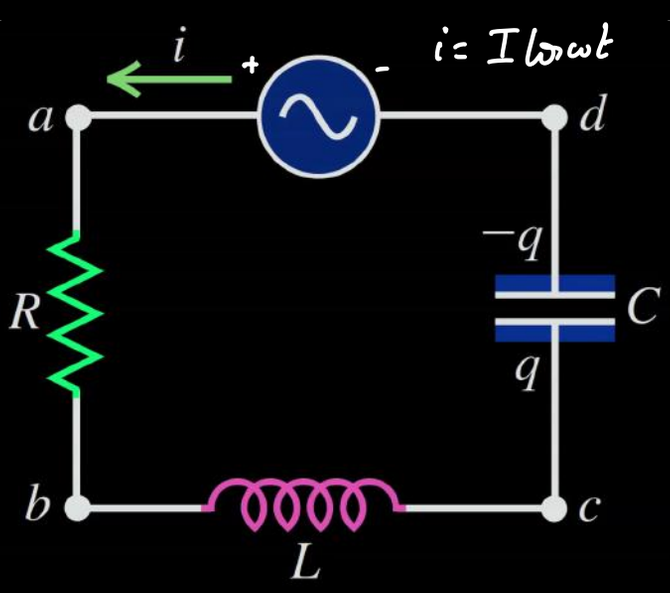
Now $\tilde{z} = 7.2j + (-20j \,||\, 15) = 7.2j + (9.6 - 7.2j) = 9.6\ \Omega$

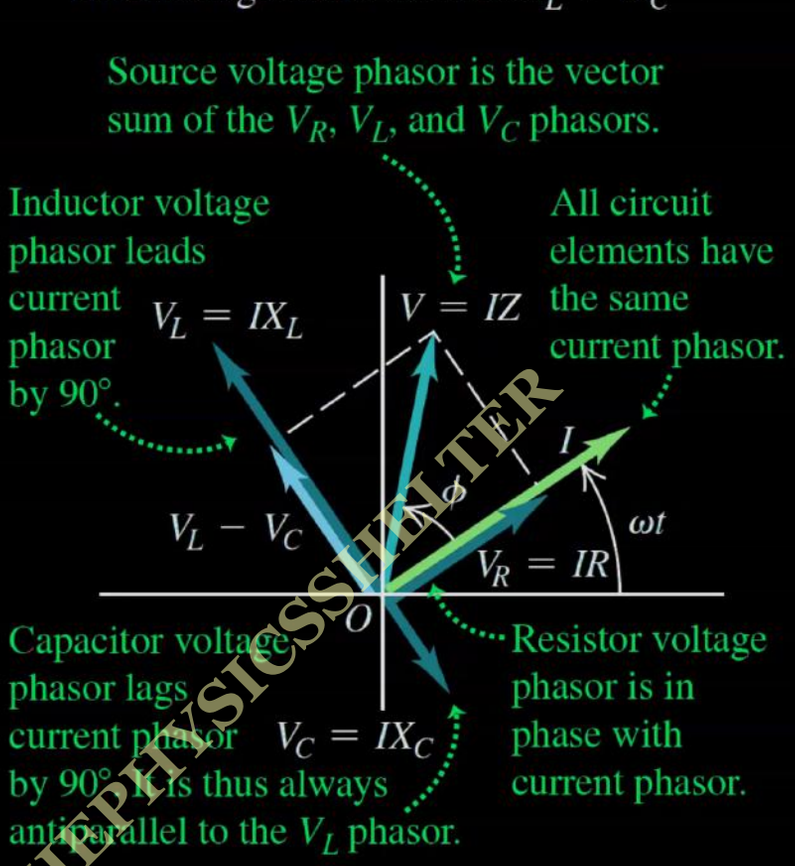
Now $\tilde{v}_c = \left(\frac{\tilde{z}_{RC}}{\tilde{z}}\right)\tilde{v}$, $\tilde{z}_{RC} = 9.6 - 7.2j = 12\angle -37°$

$$\Rightarrow \tilde{v}_c = \frac{12\angle -37°}{9.6} \times 10\angle -45° = 12.5\angle -82° \Rightarrow \tilde{v}_c = 12.5\cos(4t - 82°) \quad \text{Ans.}$$

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Series LCR Circuit Using Phasor Diagram:





The adjacent figure shows the phasor diagram for the circuit with current $i = i_0 \sin \omega t$ and $X_L > X_C$. At a general instant current phasor makes an angle ωt with horizontal axis. At any general instant voltage in inductor leads the current by a phase of $\pi/2$ and has a phasor length IX_L, similarly, for capacitor, the voltage phasor lags behind by a phase of $\pi/2$. Thus capacitor and inductor voltage phasors are 180° opposite to each other and upon phasor addition can be considered together as a phasor of length $I(X_L - X_C)$ if $X_L > X_C$ (i.e $V_L - V_C$). In the resistor, voltage and current are in same phase and length of the voltage phasor is IR. Now total voltage phasor can be obtained by taking phasor sum of all the voltages, thus, net voltage phasor is described as:

$$V = \sqrt{(V_L - V_C)^2 + V_R^2} = \sqrt{I(X_L - X_C)^2 + (IR)^2}$$

$$\Rightarrow \boxed{V = I\sqrt{(X_L - X_C)^2 + R^2}} = IZ \quad \text{where } Z = \sqrt{(X_L - X_C)^2 + R^2}$$

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which is the impedance of the circuit (i.e mod of complex impedance). Also ϕ is the angle with which voltage leads the current and, $\tan\phi = (V_L - V_C)/V_R$ thus:

$$\tan\phi = \frac{X_L - X_C}{R} \Rightarrow \boxed{\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)}$$

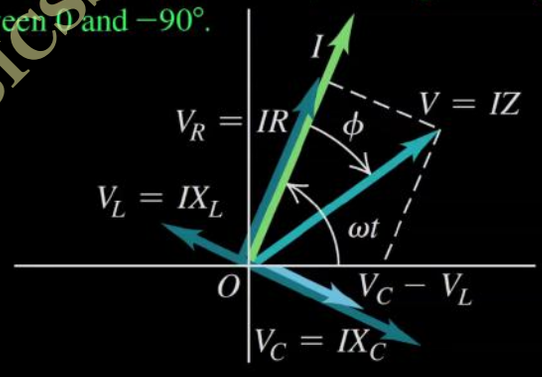
Thus knowing the ϕ and V_0, we can write the complete voltage function as:

$$\boxed{V = IZ\cos(\omega t + \phi).}$$

Similarly for the case when $X_L < X_C$, the V_C will dominate the V_L phasor and the two can be combined to give a net $V_C - V_L$ phasor which is lagging behind by a phase of $\pi/2$ relative to the current phasor in the circuit. Thus here net voltage phasor lags behind the current phasor. Here,

Phasor diagram for the case $X_L < X_C$

If $X_L < X_C$, the source voltage phasor lags the current phasor, $X < 0$, and ϕ is a negative angle between 0 and $-90°$.



$$\tan\phi = (V_C - V_L)/R = (X_C - X_L)/R$$

$\Rightarrow \phi = \tan^{-1}(X_C - X_L)/R$, just like the previous case, here also,

$$V = \sqrt{(V_C - V_L)^2 + R^2} = I\sqrt{(X_C - X_L)^2 + R^2} = IZ.$$

Now total voltage is given as $\boxed{V = IZ\cos(\omega t - \phi)}$

→ Here $Z = \sqrt{\left(\omega L - \frac{1}{\omega C}\right)^2 + R^2}$

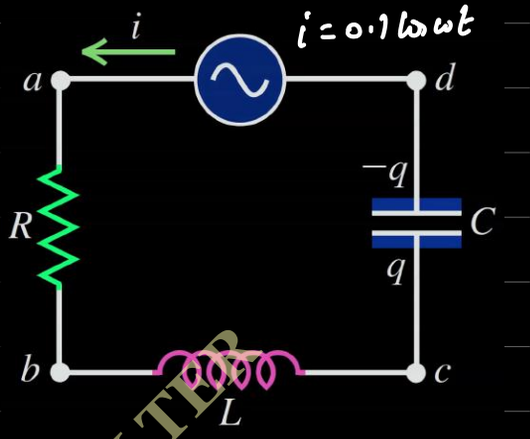
→ Here ϕ is the angle by which voltage lags behind the current or we can say the angle by which current leads the voltage, thus if instead of current, the voltage function were given as $V = V_0\cos(\omega t)$ then $i(t)$ would be simply $i = \frac{V_0}{Z}\cos(\omega t + \phi)$.

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→ Since $V = IZ$ holds for peak voltages and currents, it also holds for rms voltages and currents because $V_{rms} = V/\sqrt{2}$ and $I_{rms} = I/\sqrt{2}$. Thus $V_{rms} = I_{rms} \times Z$.

Example:

In the series circuit of adjacent figure, suppose $R = 300\ \Omega$, $L = 60$ mH, $C = 0.50\ \mu$F, , and $\omega = 10{,}000$ rad/s. Find the reactances X_L and X_C, the impedance Z, the voltage amplitude V, the phase angle φ, and the voltage amplitude across each circuit element. Also find expressions for the time dependence of the the instantaneous voltages across the resistor (v_R), inductor (v_L), capacitor (v_C) and ac source (v).



Solution:

$X_L = \omega L = 10000 \times 60 \times 10^{-3} = 600\ \Omega$ Ans

$X_C = 1/\omega C = 1/(10{,}000 \times 0.5 \times 10^{-6}) = 200\ \Omega$ Ans.

$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{(300)^2 + (400)^2} = 500\ \Omega$ Ans

$V = IZ = 0.1 \times 500 = 50$ V Ans.

$V_L = IX_L = 0.1 \times 600 = 60$ V Ans

$V_C = IX_C = 0.1 \times 200 = 20$ V Ans

$V_R = IR = 0.1 \times 300 = 30$ V Ans.

Now, $v_R = V_R \cos(\omega t) = 30 \cos \omega t$ Ans

$v_C = V_C \cos(\omega t - \pi/2) = 20 \cos(\omega t - \pi/2)$ Ans

$v_L = V_L \cos(\omega t + \pi/2) = 60 \cos(\omega t + \pi/2)$ Ans.

Now $X_L > X_C \Rightarrow$ circuit is inductive i.e overall voltage leads the current. $\tan\phi = (V_L - V_C)/V_R = (60 - 20)/30 = 4/3 \Rightarrow \phi = 53°$.

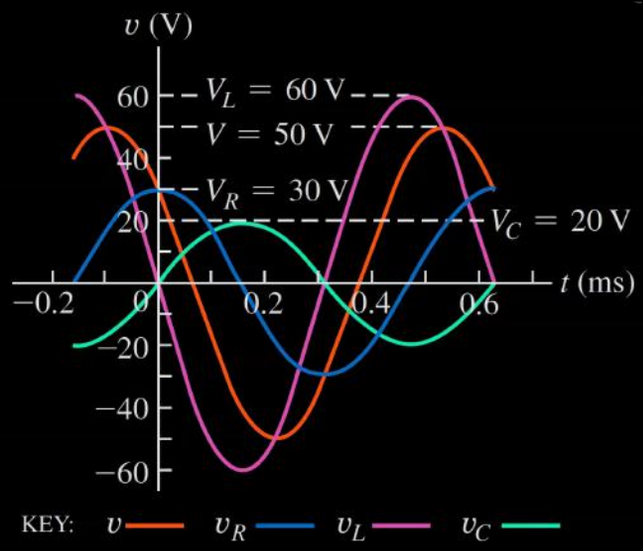
Now $v = V \cos(\omega t + \phi) = 50 \cos(\omega t + 53°)$ Ans.

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A plot of instantaneous voltages is shown in the adjacent figure. Note that

$V \neq V_L + V_R + V_C$ but $v = v_L + v_R + v_C$

i.e instantaneous total voltage is algebraic sum of instantaneous voltages across elements, though phasors amplitudes have to be added vectorially i.e $\vec{V} = \vec{V_L} + \vec{V_C} + \vec{V_R}$.



Power In An AC Circuit:

Suppose we have an A.C. circuit in which driven current and voltage differ by a phase of ϕ then, instantaneous power:

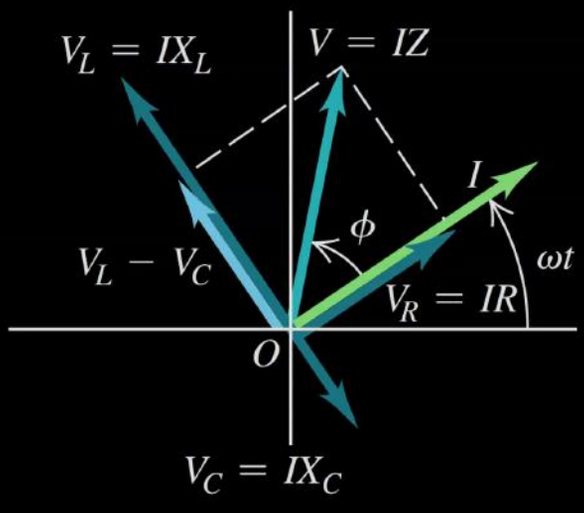
$$p = vi = V\cos(\omega t + \phi)\, I \cos\omega t$$

$$= V[\cos\omega t\cos\phi - \sin\omega t \sin\phi)\, I\cos\omega t = VI\cos^2\omega t\cos\phi - VI\sin\phi\sin\omega t\cos\omega t.$$

Now $\langle \sin\omega t\cos\omega t\rangle = \frac{1}{2}\langle \sin 2\omega t\rangle = 0$ and $\langle\cos^2\omega t\rangle = \frac{1}{2}$

$$\Rightarrow P_{avg} = \frac{VI}{2}\cos\phi = \frac{V}{\sqrt{2}} \times \frac{I}{\sqrt{2}} \times \cos\phi = V_{rms}\, i_{rms}\cos\phi$$

Thus $\boxed{P_{avg} = V_{rms}\, i_{rms}\cos\phi}$



→ For a simple series LCR circuit,

$$\cos\phi = \frac{V_R}{V} = \frac{IR}{Iz} = \frac{R}{Z}$$

Thus $\boxed{\cos\phi = R/Z}$

→ $P_{avg} = (i_{rms} \times Z) \times (i_{rms}) \times \frac{R}{Z}$

$\Rightarrow$ $\boxed{P_{avg} = i_{rms}^2 R}$

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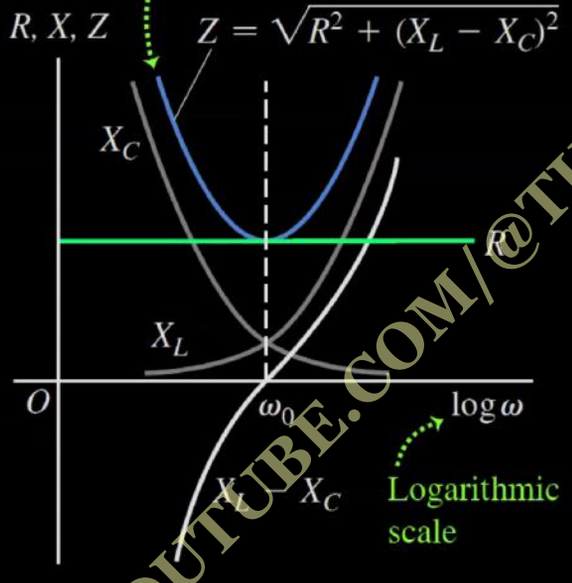
→ The terms $\cos\phi$ is known as power factor of the circuit.

→ In general we can also talk about power factor of a segment of a circuit rather than power factor of entire circuit. In such cases, if impedance $\tilde{Z} = R + jx$ then power factor is simply

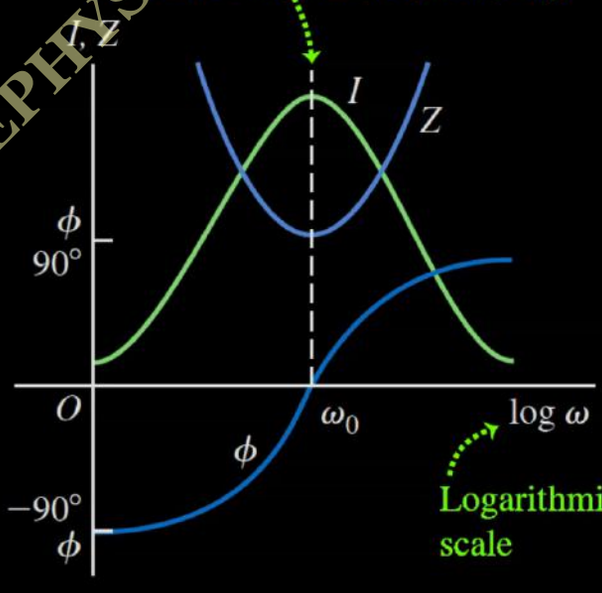
$$\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + x^2}}$$

Impedance And Phase Variation In Series A.C. Circuit:

Reactance, resistance, and impedance as functions of angular frequency



Impedance, current, and phase angle as functions of angular frequency



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The curves in the above figures show the variation of reactance and impedance with driving frequency and also the phase angle variation between current and source voltage for a series LCR circuit.

→ Recall that $x_c = 1/\omega C$, thus at very low driving frequencies, capacitive reactance becomes very large and the effect of resistance and inductance become relatively smaller. Thus at very low frequencies the circuit is said to be capacitive (i.e Voltage lags behind the source current by nearly 90°).

→ At very high frequencies, the inductive reactance ($X_L = \omega L$) becomes very high and the capacitive reactance ($1/\omega C$) is very low. Thus effects of resistance and capacitance are dominated by the inductor and the circuit is said to be inductive. The voltage leads the source current by a phase of nearly 90°.

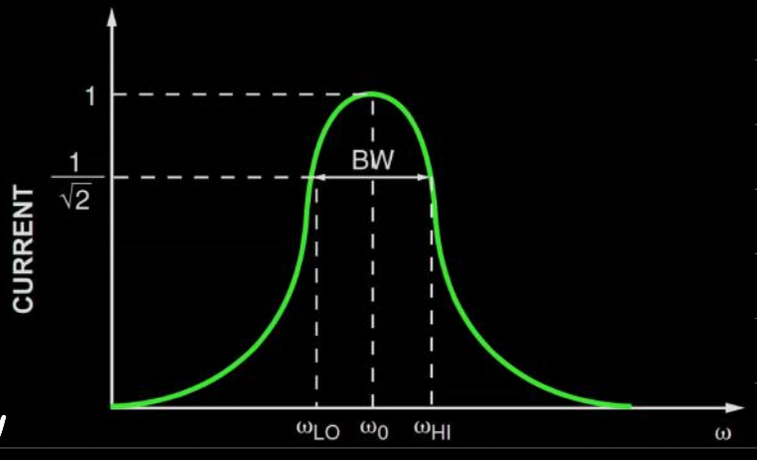
→ $Z = \sqrt{(X_L - X_C)^2 + R^2}$, this impedance is minimum when $X_L = X_C$ i.e $\omega L = 1/\omega C$ or $\omega = 1/\sqrt{LC}$. At this frequency, the current is maximum and in phase with source voltage, this condition is called resonance. Thus at resonance:

$$\boxed{\omega_0 = 1/\sqrt{LC}} \qquad \boxed{I_{res} = I_{max} = V/R} \;, \qquad \boxed{Z = R}$$

$\boxed{\cos\phi = 1}$. In general resonance in A.C. circuits means that $\tilde{Z}$ is real.

HALF POWER FREQUENCIES, BAND WIDTH AND QUALITY FACTOR:

In previous section we saw that maximum current in a series LCR circuit occurs when $\omega = \omega_0 = 1/\sqrt{LC}$. With this condition, the power dissipated in the circuit is also maximum on an average. When the frequency is increased or decreased below this number the power decreases.



For purpose of certain electrical and communication engineering applications, the frequencies at which the power becomes half are of special interest. Now $P_{max} = I_{max}^2 R$, For half power, Current must decrease to $I/\sqrt{2}$ ($\because P \propto I^2$). This requires that impedance must increase $\sqrt{2}$ fold. We can use this to calculate half power frequencies. At resonance, $Z = R$, thus at half power, $|Z| = R\sqrt{2}$

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$$\Rightarrow \sqrt{(X_C - X_L)^2 + R^2} = R\sqrt{2} \quad \Rightarrow \left(\omega L - \frac{1}{\omega C}\right)^2 = R^2$$

$\Rightarrow \omega L - 1/\omega C = \pm R$, we are interested in only positive roots of these equations,

$$\omega_{L0} = \frac{-R + \sqrt{R^2 + 4\omega_0^2 L^2}}{2L}$$

$$\omega_{H1} = \frac{R + \sqrt{R^2 + 4\omega_0^2 L^2}}{2L}$$

→ Here $\omega_0 = 1/\sqrt{LC}$ i.e resonance frequency.

BANDWIDTH: The difference of half power angular frequencies is called Bandwidth. Thus, $B.W = \Delta\omega = R/L$

$$B.W = \Delta\omega = R/L$$

→ It is worth remembering that $\omega_0 = \sqrt{\omega_{H1} \times \omega_{L0}}$

→ If $(R/2L) << \omega_0$ (which is often the case in actual communication circuits) then we can approximately write, $\omega_{H1} = \omega_0 + R/2L$ and $\omega_{L0} \approx \omega_0 - R/2L$ i.e half power frequencies may be considered equidistant from resonance frequency.

QUALITY FACTOR: The ratio $\omega_0/\Delta\omega$ is called quality factor. (This is because if this factor is large, the resonance peak is very sharp). Thus $Q = \frac{1}{\sqrt{LC}} \Big/ (R/L) = \frac{1}{\sqrt{LC}} \times \frac{L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$

Thus: $$Q = \frac{\omega_0}{\Delta\omega} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

→ Quality factor is dimensionless.

→ R/L has dimensions of T^{-1}

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EXAMPLE: Show that powers at $\omega_1 = \eta\omega_0$ and $\omega_2 = \omega_0/\eta$ is same.

Solution: Let let at resonance $x_L = x_C = x$ then at $\omega_1 = \eta\omega_0$, $x_L = \eta x$ $x_C = x/\eta$ ($\because x_L = \omega L$, $x_C = 1/\omega C$) thus,

$$Z_{\omega_1} = \sqrt{(\eta x - x/\eta)^2 + R^2}$$, Now with $\omega_2 = \omega_0/\eta$,

$x_L = x/\eta$ and $x_C = \eta x$ thus, $Z_{\omega_2} = \sqrt{(x/\eta - \eta x)^2 + R^2}$

thus we can see that $Z_{\omega_1} = Z_{\omega_2}$ $\therefore$ $I_{\omega_1} = I_{\omega_2}$ thus, the powers at both frequencies are same.

EXAMPLE: SBT: 21:

A series RLC circuit is in resonance with a source of frequency $\omega_0 = 10$ MHz. The current amplitude in the circuit is I_0. It was found that when a different source of frequency $\omega = \omega_0 + \Delta\omega$, [$\Delta\omega = 10$ KHz] was used the current amplitude in the circuit was only 1% of I_0. Find the inductance in the circuit if it is known that resistance in the circuit is $R = 0.314\ \Omega$.

Solution: $\omega + \Delta\omega = 1.001\,\omega$, now for current to become 1%, Z must increase 100 times. Now let x be the reactance at resonance, then:

$$\sqrt{\left(1.001\,x - \frac{x}{1.001}\right)^2 + R^2} = 100R$$

$$\Rightarrow (0.002\,\omega_0 L)^2 + R^2 = 10000R^2 \Rightarrow L = \frac{\sqrt{9999}\,R}{0.002\,\omega_0}$$

$$\Rightarrow L \approx \frac{100 \times \pi/10}{0.002 \times 2\pi \times 10^7} = 0.25\ \text{mH}$$ Ans.

EXAMPLE: SBT: 13:

A box has a large electric circuit inside it. When it was connected to an ac generator it was found that it was putting a lot of load on the generator and the power factor of the box was $1/\sqrt{2}$. A capacitor of capacitance C was connected in series with the box and the power factor of the circuit became equal to the ideal value. Find the impedance of the box. The generator has an angular frequency of ω.

Solution: $\cos\phi = R/Z$ $\Rightarrow$ initially, $Z = \sqrt{2}R$ $\Rightarrow$ $\tilde{Z} = R + jR$

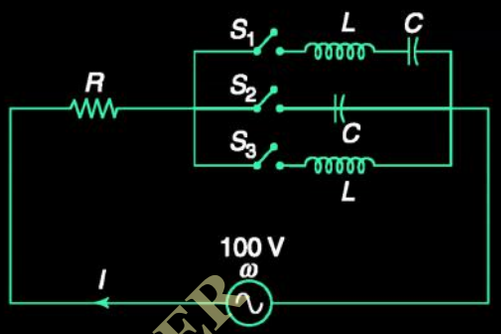
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Finally $R/z = 1 \Rightarrow R/|R + jR - jX_c| = 1$

$\Rightarrow R = X_c = \frac{1}{\omega C}$ thus initial $z = \sqrt{2}R = \sqrt{2}/\omega C$ ohm.

EXAMPLE: SBT: 22:

In the circuit shown in the figure, one of the three switches is kept closed and other two are open. The value of resistance is R = 20 Ω. When the angular frequency(ω) of the 100 V source is adjusted to 500 rad/s, 1000 rad/s and 2000 rad/s it was found that the current I was 4A, 5A and 4A respectively. (a) Which switch is closed? (S1, S2 or S3) (b) Find the value of L and C.



Solution: If there were only X_L or X_c in the circuit, impedance would vary monotonically with ω. Thus we need to have both X_L and X_c therefore, S_1 must have been closed. Also since current is same for the frequencies 1000/2 and 1000 x 2 therefore $\omega_0 = 1000$ Rad/sec. Now let inductive reactance at resonance be x. Now ∵ current becomes 4/5 times at $\omega_0/2$, impedance at $\omega_0/2$ must be $\frac{5}{4}R$. Thus:

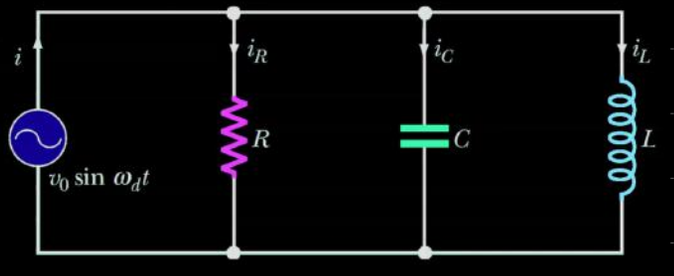
$$\sqrt{(X/2 - 2x)^2 + R^2} = 5/4\,R \Rightarrow x = R/2 = \omega_0 L = \frac{1}{\omega_0 C}$$

$\Rightarrow L = R/2\omega_0 = 20/2 \times 1000 = 10$ mH Ans.

$C = 2/\omega_0 R = 2/(1000 \times 20) = 100\ \mu F$ Ans.

EXAMPLE:

For circuit of the Figure, find the instantaneous current through each element. Also find (a) the total instantaneous current through the source and (b) the expression for the phase angle of this current and the impedance of the circuit. Draw a phasor diagram to illustrate the relationship of the instantaneous current to the instantaneous emf.



Solution: $i_R = V/R = \frac{V_0}{R} \sin \omega t$. Now in capacitor, current leads the voltage by π/2, thus $i_c = \frac{V_0}{X_c} \sin(\omega t + \pi/2) = V_0 \omega C \sin(\omega t + \pi/2)$

In inductor current lags behind by π/2 thus $i_L = \frac{V_0}{\omega L} \sin(\omega t - \pi/2)$

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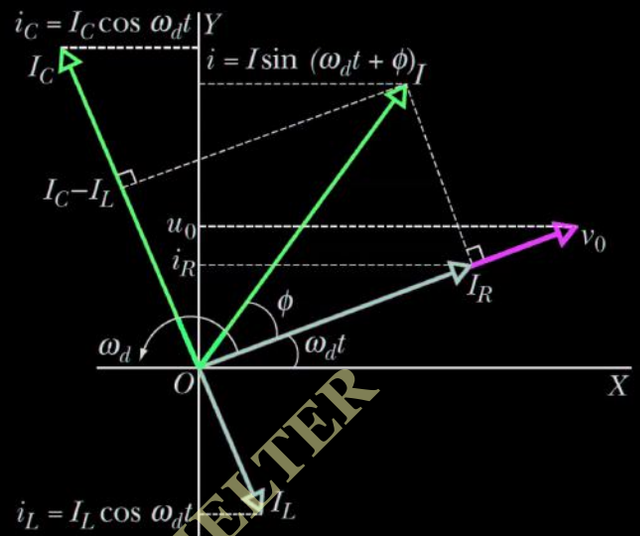
Thus $i = V_0\left[\frac{1}{R}\sin\omega t + \left(\frac{1}{X_C} - \frac{1}{X_L}\right)\cos\omega t\right] = I\sin(\omega t + \phi)$

where $I = V/Z$.

The phasor diagram for the situation is shown in adjacent figure. Here

$$I = \sqrt{\left(\frac{V_0}{X_C} - \frac{V_0}{X_L}\right)^2 + \left(\frac{V_0}{R}\right)^2} = \frac{V_0}{Z}$$

$$\Rightarrow Z = \left[\left(\frac{1}{X_C} - \frac{1}{X_L}\right)^2 + \left(\frac{1}{R}\right)^2\right]^{-1/2} \text{ Ans.}$$



Also $\tan\phi = (I_C - I_L)/I_R$

$\Rightarrow \tan\phi = (1/X_C - 1/X_L)/(1/R)$ Ans.

Method 2: Complex Numbers: $\tilde{Y} = 1/R + \tilde{B}_C + \tilde{B}_L = G + jB$

$$\Rightarrow \tilde{Y} = \frac{1}{R} + j\omega C - \frac{j}{\omega L} = \frac{1}{R} + j/X_C - j/X_L$$

Now $\tilde{I} = \tilde{V}\tilde{Y} = (V_0\angle -\pi/2) \times \left[\frac{1}{R} + j\left[\frac{1}{X_C} - \frac{1}{X_L}\right]\right]$

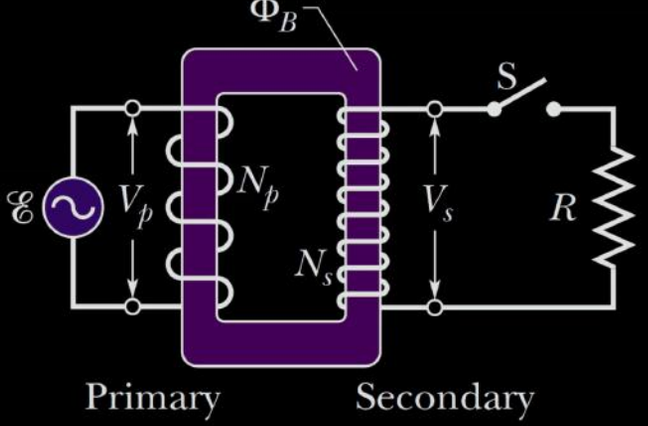
Thus $I = \frac{V_0}{Z}\cos\left(\omega t - \pi/2 + \tan^{-1}\left[(1/X_C - 1/X_L)/(1/R)\right]^{-1}\right)$

where $Z = \left[\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_C} - \frac{1}{X_L}\right)^2\right]$ Ans.

→ Resonance in LC circuit minimizes the current. (Or maximizes voltage)

TRANSFORMERS:

A transformer is a device which is used to increase or decrease the voltage at which power is being supplied. It essentially consists of two coils wound on a ferromagnetic core as shown in the figure. Ferromagnetic core increases the magnetic field several times and also ensures that flux per turn is nearly same for the primary and secondary coils.



An ideal transformer

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Now the voltage across the coils is directly proportional to the flux linkage, which is turn is proportional to the number of turns thus the ratio of primary and secondary voltages is given by,

$$\frac{V_P}{V_S} = \frac{N_P}{N_S}$$

N_P: No of turns in primary coil
N_S: No of turns in secondary coil.

In an ideal transformer there is no energy loss and the inductance are very high. Under such circumstances it can be shown using energy considerations that:

$$V_P I_P = V_S I_S \Rightarrow \frac{I_P}{I_S} = \frac{V_S}{V_P} = \frac{N_S}{N_P}$$

Thus $\frac{I_P}{I_S} = \frac{N_S}{N_P}$, V_P/I_P is sometimes called effective resistance as seen by the primary.

In an ideal transformer, losses are assumed to be zero however in a real transformer, following losses occur:

Copper Losses: The copper windings in transformers do have a finite resistance, thus due to alternating current flow there is i^2R loss occuring in them, this is known as copper loss.

Iron Loss: Due to alternating magnetic field in the ferromagnetic core, there are eddy currents produced in its matrix. The heat loss occuring due to these eddy currents is called Iron loss.

Flux Leakage Loss: In an ideal transformer we assume all field lines to be confined in the ferromagnetic core, however actually there are magnetic field lines in the surrounding space, and time varying magnetic fields causes electromagnetic waves in the space, thus leading to losses in form of field energy.

→ Besides this there are also hysteresis losses and humming losses (due to vibrations).

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Transformer Efficiency:

$$\eta = \frac{V_p I_p - \text{Iron Loss} - \text{Copper Loss}}{V_p I_p}$$

Typically other losses are much smaller than iron and copper losses, however if they are significant, they must be subtracted from the numerator.

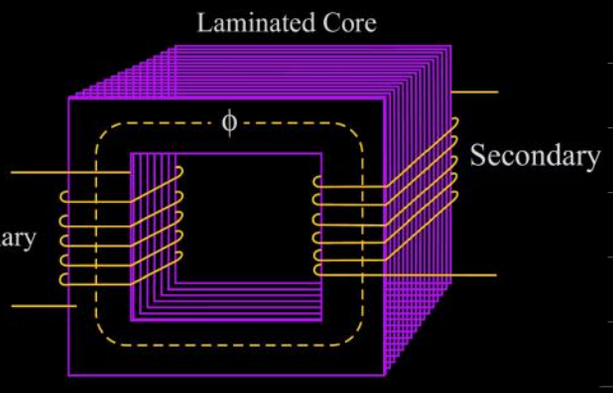
→ Strictly speaking power should be written as $VI\cos\phi$, however for JEE purposes we shall only write VI (This is not strictly correct, but just a working method for JEE problems), a detailed treatment on transformer theory is done in electrical machines text books, however, for JEE purposes we will deal with only certain specific calculations done in set specific ways.

→ If $N_s > N_p$ then $V_s > V_p$ such a transformer is called step-up transformer.

→ If $N_s < N_p$ then $V_s < V_p$, such a transformer is called step-down transformer.

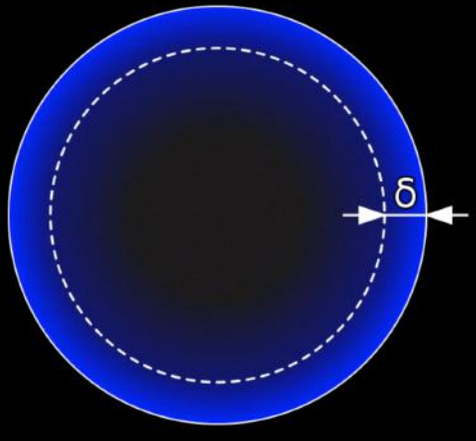
→ If $N_s = N_p$, such a transformer is called an isolator.

→ In Order to reduce iron losses, transformer core is made of laminated sheets rather than a single solid iron block. This limits the paths available for eddy currents and hence reduces the iron losses.



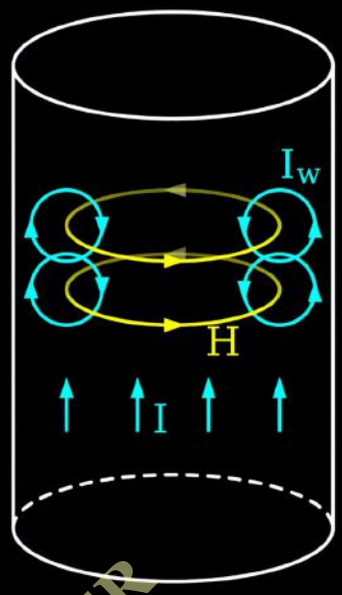
Skin Effect:

When alternating current flows in a conductor, the current density is more near the surface, while less near the centre of the conductor. This is known a skin effect and it gets more and more pronounced at higher frequencies. This effect occurs due to self induction



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within a current carrying wire. This is illustrated in the adjacent figure. If the current is increasing with time in upward direction, the induced EMF is such that it opposes the current near the centre and supports the current near the periphery thus increasing the current density near the surface. Skin effect thus increases the effective resistance of conductors.



EXAMPLE:

The core of a 100-kVA, 11000/550 V, 50-Hz, Single Phase, core type transformer has a cross-section of 20 cm × 20 cm. Find (i) the number of H.V. and L.V. turns per phase and (ii) the e.m.f. per turn if the maximum core density is not to exceed 1.3 Tesla. Assume a stacking factor of 0.9. [Stacking factor is the fraction of cross section occupied by the ~~insulating lamination~~ iron between the core sheets]

Solution: $V_{p_{rms}} = (N_p \times B \times A \times \omega \times \eta)/\sqrt{2}$, $\eta \rightarrow$ stacking factor. ($\because \phi_{max} = BA\omega\eta$).

$$\Rightarrow N_p = \frac{V_p\sqrt{2}}{BA \times 2\pi f \times \eta} = \frac{11000 \times \sqrt{2}}{1.3 \times 0.04 \times 2\pi \times 50 \times 0.9} \approx 1060 \text{ turns Ans.}$$

Now $N_s = \frac{V_s}{V_p} \times N_p = \frac{550}{11000} \times 1060 = 53$ turns Ans.

Now EMF per turns $= (BA\omega\eta)/\sqrt{2} = V_p/N_p = V_s/N_s = 550/53 \approx 10.4167$ Ans.

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EXAMPLE :

A 25 kVA, single-phase transformer has 250 turns on the primary and 40 turns on the secondary winding. The primary is connected to 1500 Volt, 50 Hz mains. Calculate (i) Primary and Secondary currents on full-load, (ii) Secondary e.m.f., (iii) maximum flux in the core.

Solution: $V_p I_p = V_s I_s = P \Rightarrow I_p = P/V_p = 25\times10^3/1500 = 16.67\ A$

$V_s = \frac{N_s}{N_p} V_p = \frac{40}{250} \times 1500 = 240$ Volt, $I_s = P/V_s = 25000/240 = 104.2\ A.$

Now $V_p \times \sqrt{2} = \phi_{max} \times \omega \times N_p \Rightarrow \phi_{max} = \sqrt{2} V_p / (2\pi f N_p)$

$\Rightarrow \phi_{max} = \sqrt{2} \times 1500/(2\pi \times 50 \times 250) = 0.027\ Wb = 27\ mWb$ Ans.

EXAMPLE :

A transformer on a utility pole operates at $V_P = 8.4$ kV on the primary side and supplies electrical energy to a number of nearby houses at $V_S = 120$ V, both quantities being RMS values. The average rate of energy consumption (or dissipation) in the houses served by the transformer is 84 kW. Assume an ideal step-down transformer, a purely resistive load, and a power factor of unity.

(A) The turns ratio N_P/N_S of the transformer is 70.

(B) The current in primary and secondary coils are $I_P = 10$ A, $I_S = 700$ A.

(C) Resistance in primary and secondary coils are $R_P = 840\Omega$, $R_S = 0.17\Omega$

(D) None

Solution: $N_p/N_s = V_p/V_s = 8.4\times10^3/120 = 70$ (A) is correct

$V_s I_s = P \Rightarrow I_s = P/V_s = 84\times10^3/120 = 700\ A$

$I_p = I_s N_s/N_p = 700 \times 1/70 = 10\ A \Rightarrow$ (B) is correct.

Now resistance across secondary, $R_s = V_s/I_s = 120/700 = 0.17\ \Omega$

Resistance as seen by primary, $R_p = V_p/I_p = 8.4\times10^3/10 = 840\ \Omega$

(Thus C could be correct if it is meant as effective resistance as seen by primary or actually connected across secondary. However since we have assumed it to be a lossless transformer, actual resistances within the transformer coils are zero).

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EXAMPLE :

The overall efficiency of a transformer is 90%. The transformer is rated for an output of 9000 watts. The primary voltage is 1000 V. The ratio of turns in the primary to the secondary coil is 5: 1. The iron losses at full load are 700 watts. The primary coil has a resistance of 1 Ω.

(A) The voltage in the secondary coil is 200 V
(B) the current in the primary coil is 10 A
(C) the Joule's heating loss in the secondary coil is 200 W
(D) the resistance of the secondary coil is nearly 0.1 Ω

Solution: $P_{out} = V_s I_s = \left(V_p \times \frac{N_s}{N_p}\right) \times I_s \Rightarrow I_s = P_{out} \times \frac{N_p}{N_s} \times \frac{1}{V_p}$

$\Rightarrow I_s = 9000 \times 5 \times \frac{1}{1000} = 45\ A.$

Now $P_{out} = P_{in} \times \eta \Rightarrow P_{in} = P_{out}/\eta = 10{,}000\ W \Rightarrow$

Total loss = 1000 W $\Rightarrow$ Copper loss = 1000 W − Iron loss = 300 W.

Now $P_{in} = V_p I_p \Rightarrow I_p = 10000/1000 = 10\ Amp.$

Now copper loss in primary, $I_p^2 R_p = 100 \times 1 = 100\ W$

$\Rightarrow$ copper loss in secondary = 300 − 100 = 200 W = $I_s^2 R_s$

$\Rightarrow R_s = 200\ W / I_s^2 = 200/45^2 \approx 0.1\ \Omega$ A, B, C, D are all correct.

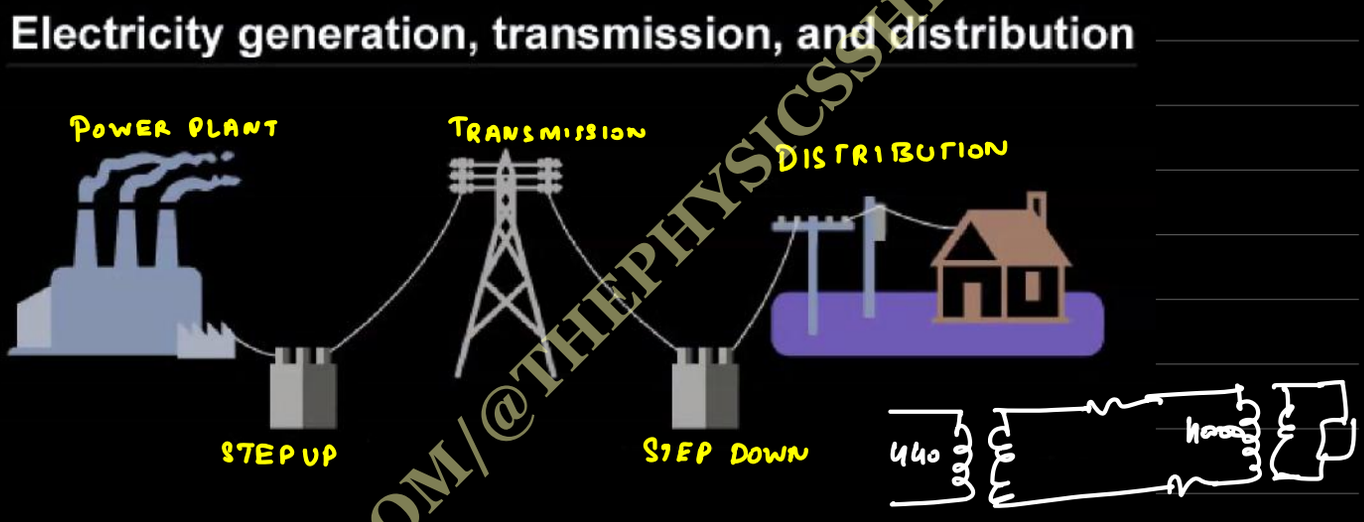
→ Note that if primary resistance were not given we could assume primary and secondary resistances in ratio of number of turns and divide copper losses in ratio $I_p^2 N_p : I_s^2 N_s$. (That gives a different answers for copper losses (60 + 240)).

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EXAMPLE:

A small town with a demand of 800 kW of electric power at 220 V is situated 15 km away from an electric plant generating power at 440 V. The resistance of the two wire line carrying power is 0.5 Ω per km. The town gets power from the line through a 4000-220 V step-down transformer at a sub-station in the town. (a) Estimate the line power loss in the form of heat. (b) How much power must the plant supply, assuming there is negligible power loss due to leakage? (c) Characterize the step up transformer at the plant.

Solution:



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Assuming lossless stepdown, $V_p I_p = V_s I_s = 800\,kW \Rightarrow I_p = \frac{800\,kW}{4000}$

$\Rightarrow I_p = 200\,A$, line resistance $R = 15 \times 0.5 \times 2 = 15\,\Omega$

$\Rightarrow$ line loss: $P_L = I_p^2 R = 200^2 \times 15 = 600\,kW$

Thus plant must supply $800\,kW + 600\,kW = 1400\,kW$.

Now potential drop along transmission line $= I_p \times R = 200 \times 15 = 3000$

$\Rightarrow$ Powerplant transformer must step up to $4000 + 3000$ Volt i.e $7000\,V$

Thus required step up transformer: $440/7000$ Volt.

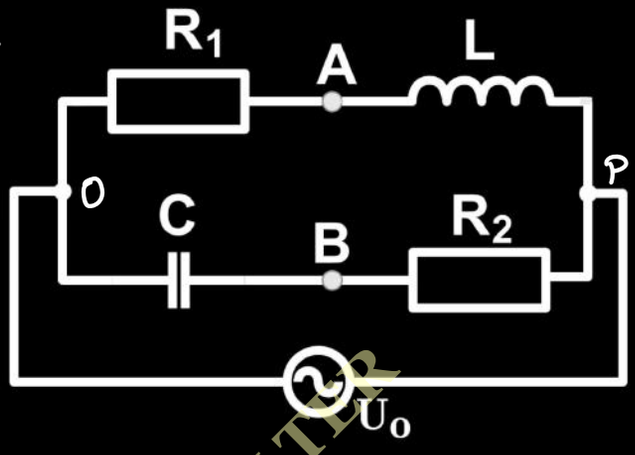
EXAMPLE: Repeat the previous problem if step down used was 40000-220.

Solution: $I_p = P_{out}/V_p = 800 \times 10^3 / 40{,}000 = 20\,A \Rightarrow$ Line loss $= 20^2 \times 15$

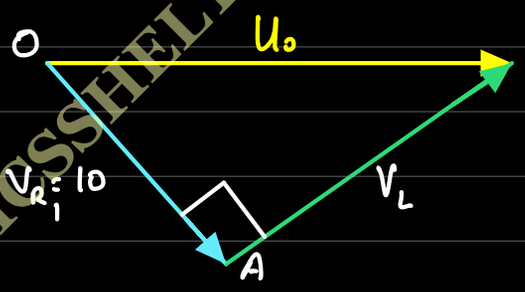
$\Rightarrow P_L = 6\,kW$, thus required supply $= 800 + 6 = 806\,kW$. Line potential drop, $V = 20 \times 15 = 300\,V \Rightarrow$ step up: $440 \rightarrow 40300$ Volt. Ans.

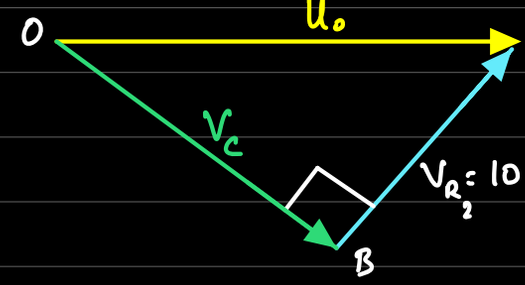
→ The above example clearly shows why long distance transmission is done at very high voltages.

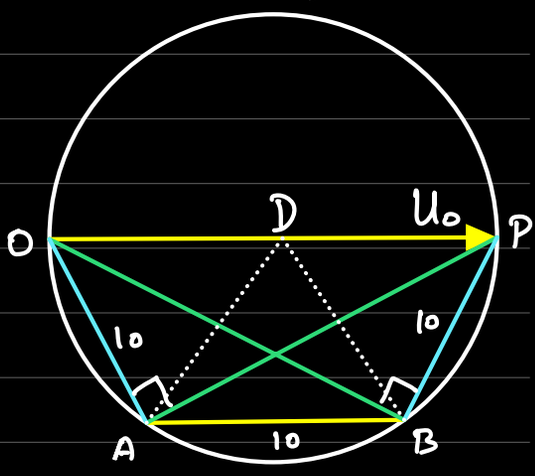
EXAMPLE : The adjacent circuit consists of a capacitor, inductance and two resistors. The voltage across both resistors is 10 V, and the voltage across A and B is also 10 Volts. Find the applied voltage U_0 in volts.



Solution: Representing U_0 by a horizontal phasor, the phasor diagrams for upper branch is as shown in the adjacent figure. Similarly the phasor diagram for lower branch is shown in the next figure. Now combining the two phasor diagrams and using the fact that angle in a semicircle is 90°, We get a geometrical construction as shown in the next figure. Now chords OA, AB and BP are each 10 Volts (AB is given as 10 Volts in the problem). Thus $\angle ODA = \angle ADB = \angle BDP = 60°$ each. Thus $\triangle ODA$ must be equilateral ∴ $OD = DP = 10$, hence applied voltage must be 20 Volts Ans.

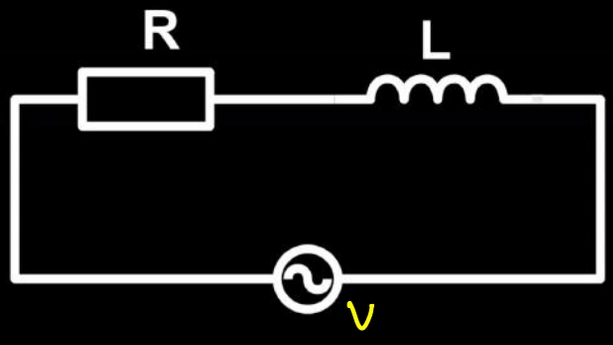






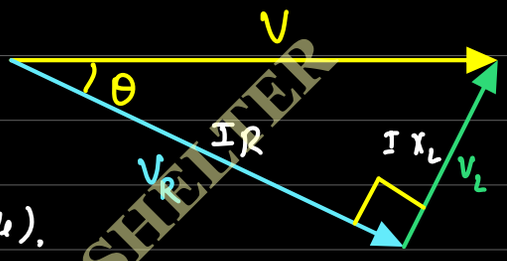
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EXAMPLE: In a series L-R circuit, it is observed that when voltage across source is maximum, the voltage across inductor is 3V and that across resistance is 4V. If resistance is 2Ω, what is the reactance of the inductor.



Solution: The phasor diagram of the LR branch at maximum V is shown in adjacent figure. According to question: $V_R = IR\cos\theta = 4$ (1) and $V_L = I x_L \sin\theta = 3$ (2) (I → Current Amplitude).

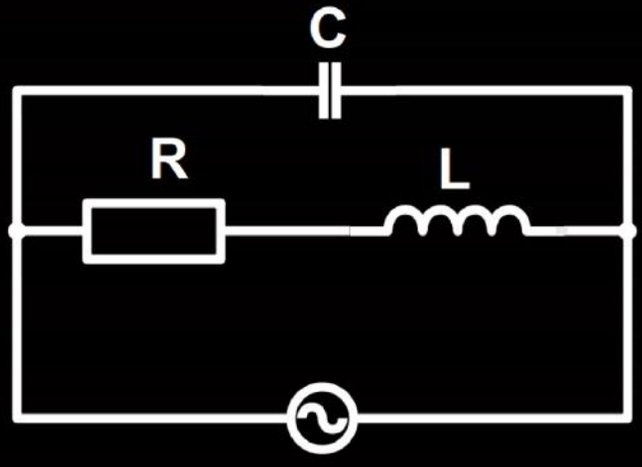
Also $V_L + V_R = 7$ (3)



Also $IR\sec\theta = 7$ (4) From (1) & (4) $I^2R^2 = 28 \Rightarrow IR = \sqrt{28}$

$\Rightarrow \sec\theta = 7/\sqrt{28}$ Now $x_L = R\tan\theta = 2 \times \frac{\sqrt{49-28}}{\sqrt{28}} = \sqrt{3}\,\Omega$ Ans.

EXAMPLE: In the shown circuit, the current is independent of choice of R. (a) Find relation between x_L and x_C (b) Find driver frequency.



Solution: If current is independent of R, for a simple analysis, we can choose $R = 0$ or $R = \infty$ and equate the impedances for both cases.

Now if $R = \infty$, $Z = x_C$ and if $R = 0$, $\tilde{Z} = \frac{-jx_C \times jx_L}{j(x_L - x_C)}$

$\Rightarrow Z = x_L x_C / (x_L - x_C)$, Now equating Z:

$$x_C = x_L x_C / (x_L - x_C) \Rightarrow \frac{x_L}{(x_L - x_C)} = 1 \Rightarrow x_L = x_C - x_L$$

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$\Rightarrow$ $X_C = 2X_L$ Ans. Also this means, $\frac{1}{\omega C} = 2\omega L$

$\Rightarrow$ $\omega = \frac{1}{\sqrt{2LC}}$ Ans.

→ Now let us verify if our result holds for a general R. Let $X_L = x$, $X_C = 2x$ and $R = \eta x$. Then $\tilde{Z} = \frac{-2jx(\eta x + jx)}{(\eta x + jx - 2jx)} = \frac{2x[1-\eta j]}{\eta - j}$

$\Rightarrow$ $Z = 2x$, so it is independent of η indeed.

EXAMPLE:

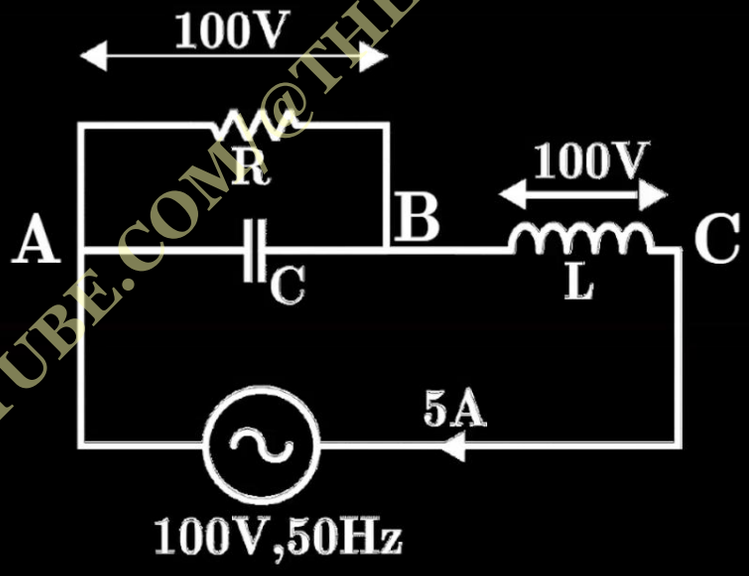
In an AC circuit shown below, voltages V_{AB} and V_{BC} are 100 V each. If the line current is 5A, then:

(A) the power consumed in the circuit is about 430W

(B) the capacitive reactance of the circuit will be 40Ω

(C) the inductance L in the circuit is 50mH

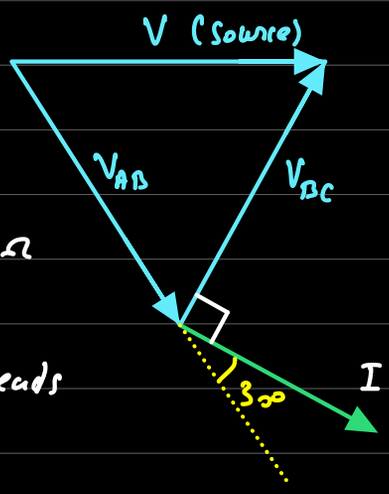
(D) the current in C is leading the voltage V_{BC} by 30°



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Solution: We will try a geometrical solution using phasors. Recognize that $V_{AB} = V_{BC}$ = Source voltage = 100 V so we expect a nice equilateral triangle from these voltages. Now in the inductor, voltage leads the current, while in RC || combination voltage lags the total current (∵ in capacitor current leads the voltage while in resistor it is in same phase).

Based on this information we come up with the adjacent phasor diagram. Also current lags behind the inductor voltage by 90°, this is shown by a green arrow. Now $X_L = 100/5 = 20\Omega$ Since for AB also V = 100 Volt and i = 5A thus, $Z_{AB} = 20\Omega$ also as clear from figure current leads the voltage AB by an angle 30°. Thus, $\tilde{Z}_{AB} = 20\angle -30°$. Thus $\tilde{Y}_{AB} = \frac{1}{20}\angle 30°$. Now let



Admittance of AB be $G + jB$ then $\sqrt{G^2+B^2} = 1/20$ and $\frac{B}{G} = \tan 30°$

or $G = B\sqrt{3}$, thus $\sqrt{4B^2} = 1/20 \Rightarrow B = 1/40$ and $G = \sqrt{3}/40$

Now Admittance of AB is also $\frac{1}{R} + \frac{j}{X_C} = \frac{\sqrt{3}}{40} + \frac{1}{40}j$

$\Rightarrow X_C = 40\Omega$ and $R = 40/\sqrt{3}\ \Omega$, thus (B) is correct.

Also $X_L = \omega L \Rightarrow L = \frac{X_L}{\omega} = \frac{20}{50 \times 2\pi} \simeq 64\,mH$ thus (C) is wrong.

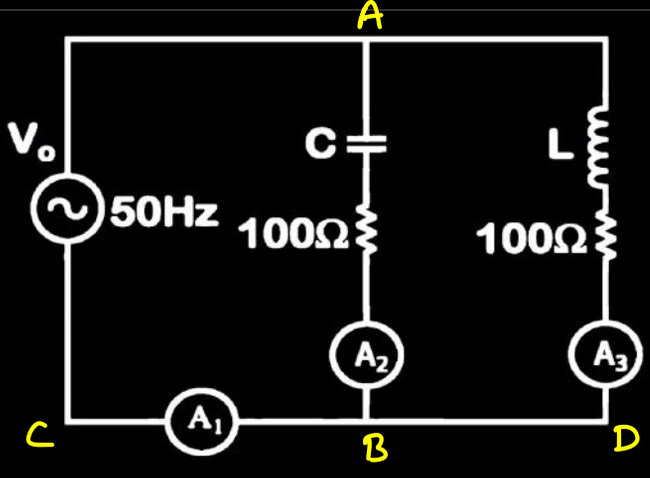
Now capacitor current is 90° anticlockwise from V_{AB}, thus lagging by 30° from V_{BC} ∴ D is wrong.

Also from phasor diagram, current lags behind the source voltage by 30°.

Thus power in the circuit $P = VI\cos\phi = 100 \times 5 \times \cos 30° = 250\sqrt{3} \simeq 433\,W$.

thus (A) is also correct.

EXAMPLE:



In the circuit shown the readings of all the three A.C. ammeters A_1, A_2 and A_3 are all equal. What are the values of L and C?

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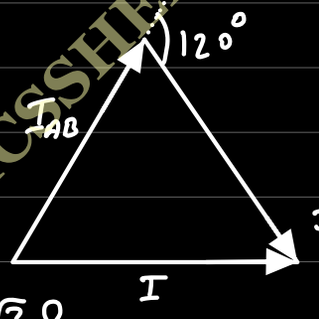
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Solution: Since currents through both AB and AD branches is same

$\therefore Z_{AB} = Z_{AD} = \sqrt{100^2 + X_L^2} = \sqrt{100^2 + X_C^2} \Rightarrow X_C = X_L = X_{say}$ ①

Thus $\tilde{Z}_{AB} = 100 - jX$, $\tilde{Z}_{AD} = 100 + jX$

Thus if current leads the voltage in Branch AD by some angle θ then it lags behind the voltage in AD by same angle θ. Let voltage be $V_0 \angle 0$, and currents in AB and AD be $I\angle\theta$ and $I\angle -\theta$. Now since readings of all three ammeters is equal, thus we will get an equilateral triangle in current phasors. Thus current in AD lags behind that in AB by $120°$, but, $I_{AD} = I\angle\theta$ and $I_{AD} = I\angle -\theta$, thus, angle between I_{AB} and I_{AD} is 2θ, therefore $2\theta = 120° \Rightarrow \theta = 60°$.



Now $\tan\theta = \frac{X}{R} = \sqrt{3} \Rightarrow X = R\sqrt{3} = 100\sqrt{3}\ \Omega$.

Thus $\omega L = 100\sqrt{3} \Rightarrow L = 100\sqrt{3}/\omega = 100\sqrt{3}/2\pi\times 50 = \sqrt{3}/\pi$ H

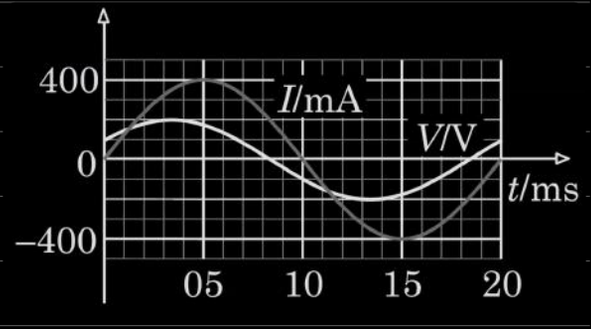
also $\frac{1}{\omega C} = 100\sqrt{3} \Rightarrow C = \frac{1}{\omega \times 100\sqrt{3}} = \frac{1}{2\pi\times 50\times 100\sqrt{3}} = \frac{10^{-4}}{\sqrt{3}\pi}$ F

$= \frac{100}{\sqrt{3}\pi}\ \mu F$ Ans.

EXAMPLE: PATHFINDER: MCQ12:

The given graph shows variation with time in the source voltage V and steady state current I drawn by a series RLC circuit. Which of the following statements is/are correct?

(a) Current lags the voltage.
(b) Resistance in the circuit is $250\sqrt{3}\ \Omega$.
(c) If capacitive reactance is 74 Ω, inductance in the circuit is approximately 560 mH.
(d) Average power dissipation in the circuit is $20\sqrt{3}$ W.



Solution: Since peak of the voltage precedes the current $\therefore$ Voltage is leading or we can say current is lading $\therefore$ option (A) is correct.

Now recognize the $i = 400\sin\omega t$ and $V = 200\sin(\omega t + 30°)$ since at $t = 0$, voltage is exactly half the peak value. Thus we figured out the lead angle of 30°.

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Now $\tilde{Z} = \tilde{V}/\tilde{i} = (200\angle -60^\circ)/(400\angle -90^\circ)\times 10^{-3} = 500\angle 30^\circ$

$\Rightarrow$

$\tilde{Z} = 250\sqrt{3} + 250j$, Thus resistance is $250\sqrt{3}$ i.e, option (B) is correct.

(c) Now $X_L - X_C = 250 \Rightarrow X_L = 250 + X_C = 324 = \omega L$

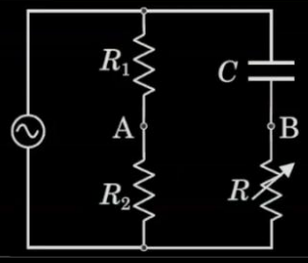
$\Rightarrow L = \frac{324}{2\pi} \times T$, from graph, $T = 20$ ms thus,

$L = \frac{324}{2\pi} \times 20\times 10^{-3} = 1031$ mH (C) is therefore incorrect

(d) Now $P = V_{rms} I_{rms} \cos\phi = \frac{400}{\sqrt{2}} \times \frac{200}{\sqrt{2}} \times \cos 30^\circ \times 10^{-3} = 20\sqrt{3}$ W

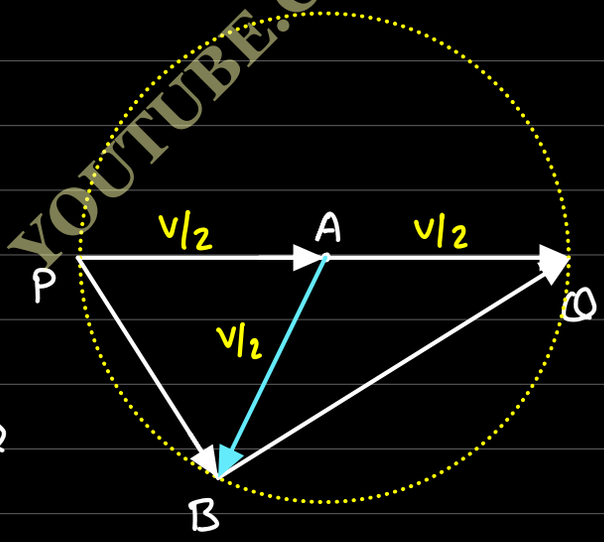
thus (D) is correct.

EXAMPLE: PATHFINDER: MCQ1S:

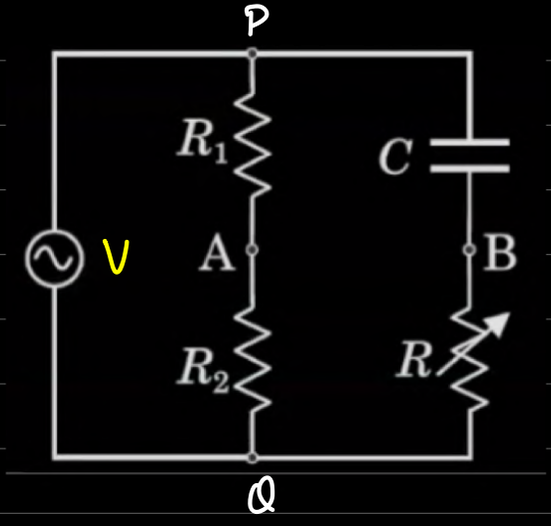


A circuit shown consists of two resistances R_1 and R_2 of equal and fixed values, a variable resistance R, a capacitor of capacitance C and an alternating voltage source. If the variable resistance is made to vary over a wide range, how does the peak voltage between the terminals A and B change?

Solution:



PA = AQ



Since resistors are identical $\therefore V_{PA} = V_{AQ} = V/2$

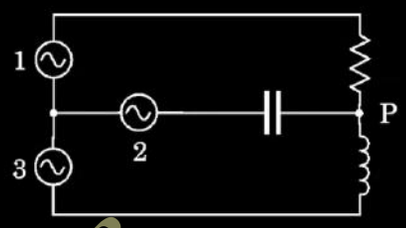
Now in capacitor branch, V_{PB} must lag behind the current by 90° and V_{BQ} must be in same phase as current. This gives the phasor diagram as shown in the figure above. $\angle PBQ$ is 90° thus

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we can visualise this triangle to be inside a semi-circle and Now as clear from the diagram, V_{AB} is simply the radius of this circle which does not change with varying R, thus V_{AB} remains constant in magnitude.

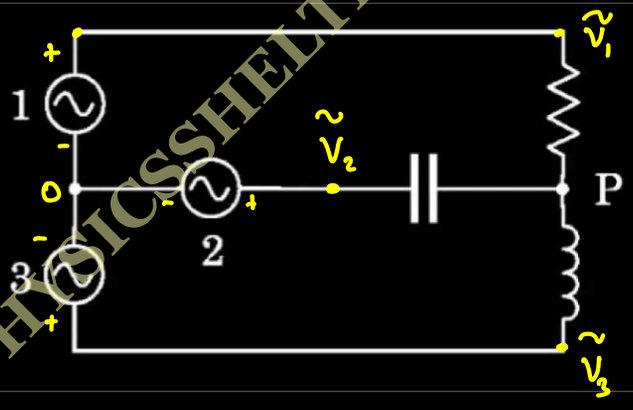
EXAMPLE: PATHFINDER : CHECK 1S:

Consider a network shown in the figure consisting of a resistance, a capacitor, an inductor and three alternating voltage sources 1, 2 and 3. Terminal voltages of the sources 1, 2 and 3 are $v_1 = V\sin(\omega t)$, $v_2 = V\sin(\omega t + 120°)$ and $v_3 = V\sin(\omega t + 240°)$ respectively and moduli of the reactances of the capacitor and the inductor are equal to the resistance. Find the voltage of the junction P.



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Solution: Whenever A.C. sources are given, polarities must be marked for proper interpretation of voltage expression, especially when there are multiple sources. Assuming polarities for expressions as shown, Applying Node Voltage at P:

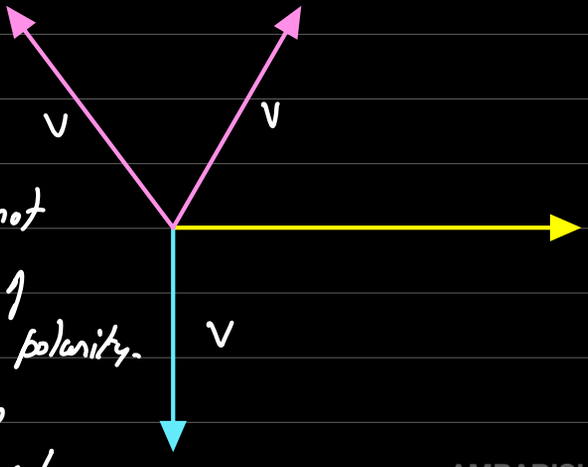


$$\frac{\tilde{V}_P - \tilde{V}_1}{R} + \frac{\tilde{V}_P - \tilde{V}_3}{jR} + \frac{\tilde{V}_P - \tilde{V}_2}{-jR} = 0$$

$$\Rightarrow \tilde{V}_P\left[\frac{1}{R}\right] = \frac{1}{R}\left[\tilde{V}_1 - j\tilde{V}_3 + j\tilde{V}_2\right]$$

$$\Rightarrow \tilde{V}_P = \tilde{V}_1 - j\tilde{V}_3 + j\tilde{V}_2, \quad \text{Now } \tilde{V}_1 = V\angle -90°,\ \tilde{V}_2 = V\angle 30°,\ \tilde{V}_3 = V\angle 150°$$

Thus $\tilde{V}_P = V\angle -90° + V\angle 60° + V\angle 120°$



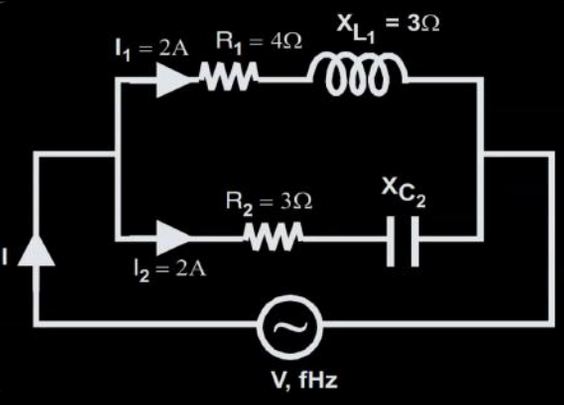
$$\Rightarrow \tilde{V}_P = V(\sqrt{3}-1)\angle 90°$$
$$= (\sqrt{3}-1)V\sin\omega t \text{ Ans.}$$

(With this polarity pathfinder Answer does not match. For Pathfinder answer polarity of V_1 must be reversed from the assumed polarity.

(Then turquoise phasor reverses and Answer becomes $\tilde{V}_P = V(\sqrt{3}+1)\angle 90° = (\sqrt{3}+1)V\sin\omega t$.

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EXAMPLE: In the shown AC circuit, the currents in two branches are shown. The capacitive reactance is unknown. Find (a) Capacitive Reactance (b) Total current supplied by the source (c) Source Voltage (d) Power factor of the circuit.

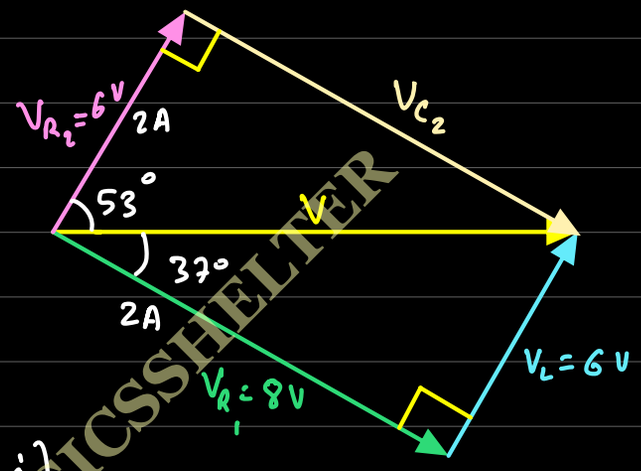


Solution: Consider the phasor diagram shown in adjacent figure. From this:

$V^2 = 8^2 + 6^2 \Rightarrow V = 10$ Volt. Ans.

Also $6^2 + V_c^2 = V^2 = 100 \Rightarrow V_c = 8$ Volt.

$\Rightarrow X_c = V_c / i_2 = 8/2 = 4\ \Omega$ Ans.



Now: $\tilde{Z} = \left[\frac{1}{4+3j} + \frac{1}{3-4j}\right]^{-1} = \frac{1}{2}[7-j]$

Thus power factor $\cos\phi = \frac{7}{\sqrt{7^2+1^2}} = \frac{7}{\sqrt{50}}$ Ans.

(Current leads voltage by ϕ since complex part is -ve (capacitive)).

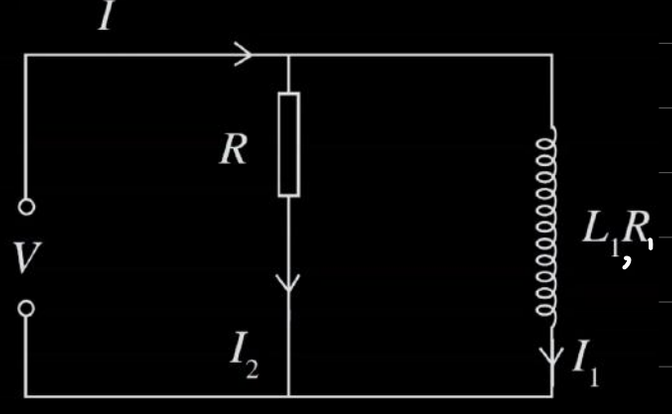
EXAMPLE: IRODOV: 4.142:

A coil and an inductance-free resistance $R = 25\ \Omega$ are connected in parallel to the ac mains. Find the heat power generated in the coil provided a current $I = 0.90$ A is drawn from the mains. The coil and the resistance R carry currents $I_1 = 0.50$ A and $I_2 = 0.60$ A respectively.

Solution: $V = I_2 R = 15$ Volt

Now for LR branch, $Z = V/I_1 = \frac{15}{0.5}$

$\Rightarrow Z_{LR} = 30\ \Omega$



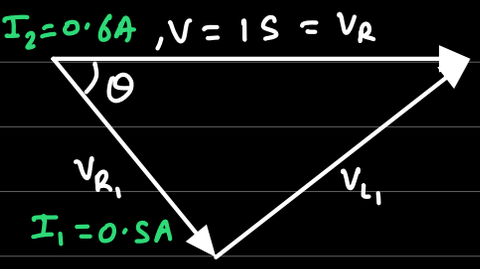
Now we draw the phasor diagram as shown in the figure on next page.

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Now currents in branches are in same phase as the voltages in resistors. Thus,

$0.5^2 + 0.6^2 + 2 \times 0.5 \times 0.6 \times \cos\theta = 0.9^2$

$\Rightarrow \cos\theta = 1/3$. Thus $V_{R_1} = V_R \cos\theta = 5$ Volt



Now Mean Power generated in coil:

$P_1 = V_{R_1} I_1 = 5 \times 0.5 = 2.5$ W Ans.

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